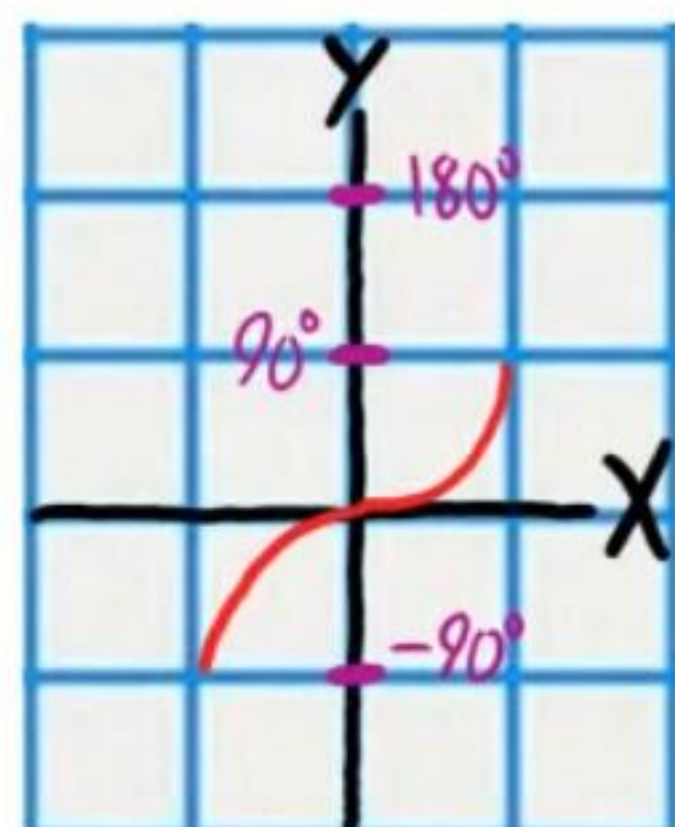


Evaluating Composite Trigonometric Functions (Part II)

Inverse Trigonometric Functions and Their Graphs

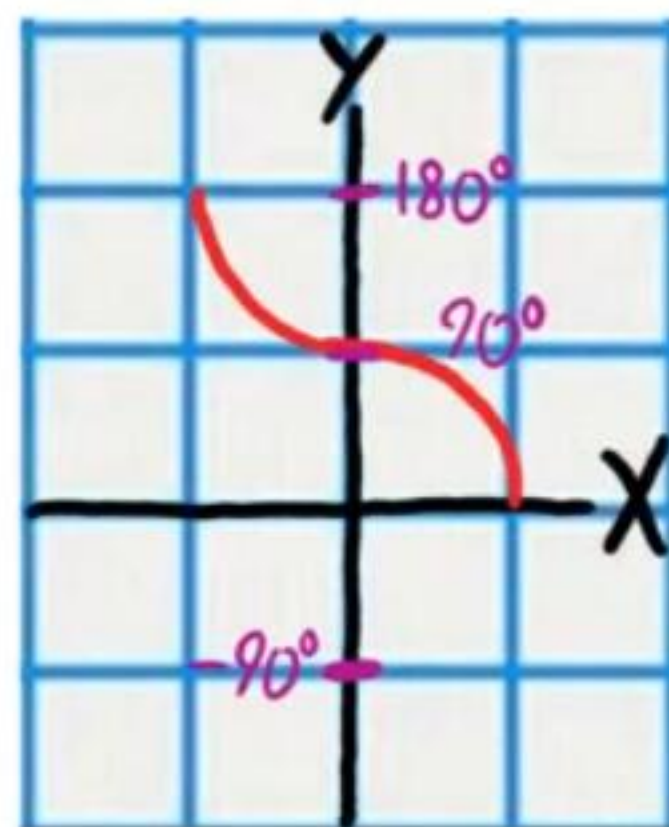
$$y = \sin^{-1}x$$

Range: $-90^\circ \leq y \leq 90^\circ$



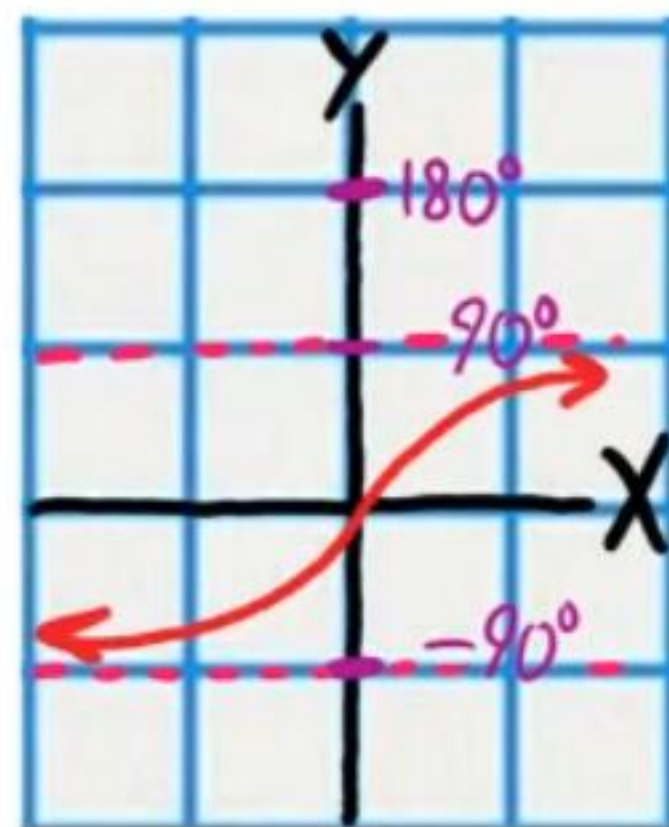
$$y = \cos^{-1}x$$

$0^\circ \leq y \leq 180^\circ$



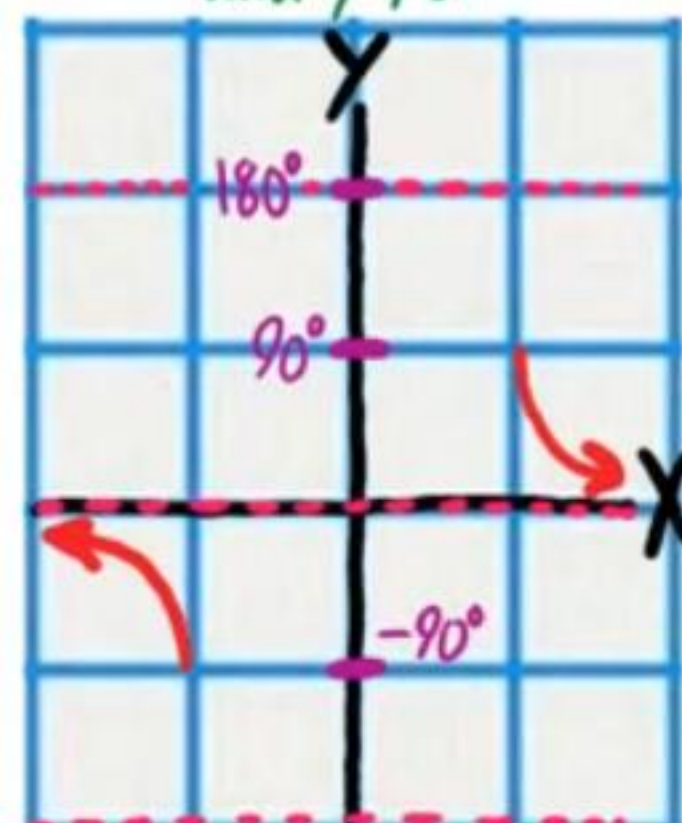
$$y = \tan^{-1}x$$

$-90^\circ < y < 90^\circ$



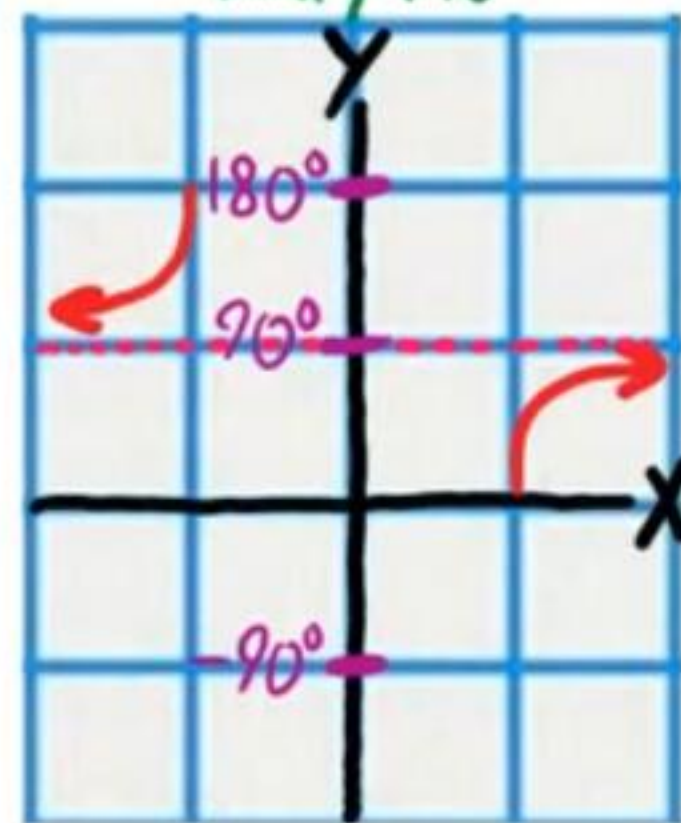
$$y = \csc^{-1}x$$

Range: $-90^\circ \leq y \leq 90^\circ$
and $y \neq 0$.



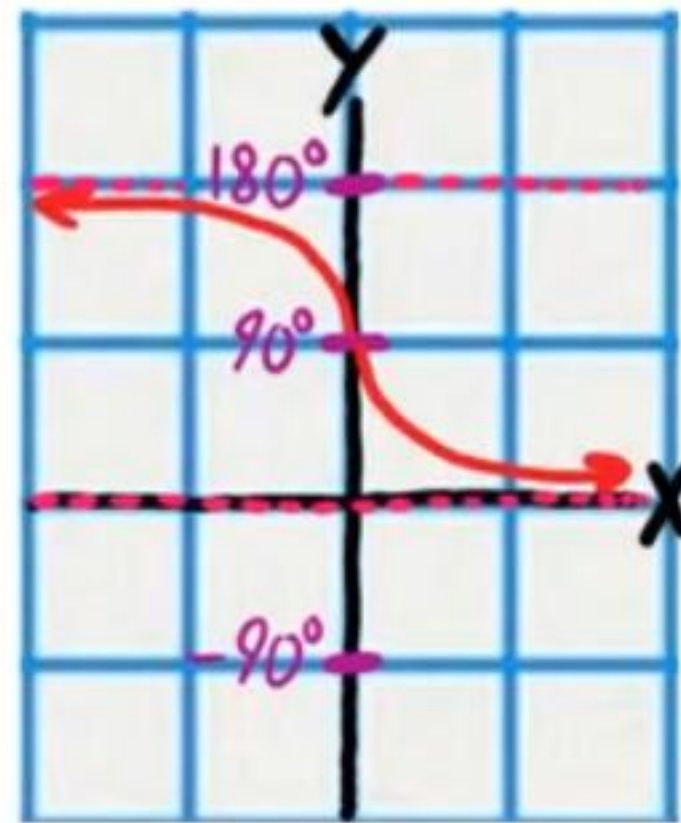
$$y = \sec^{-1}x$$

$0^\circ \leq y \leq 180^\circ$
and $y \neq 90^\circ$



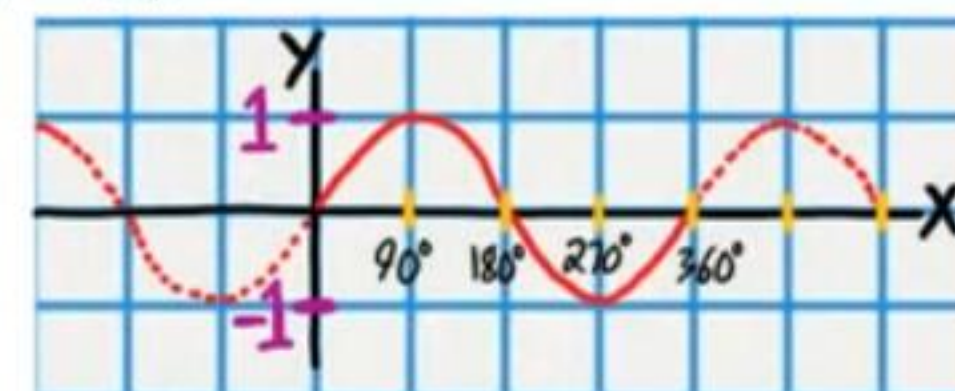
$$y = \cot^{-1}x$$

$0^\circ < y < 180^\circ$

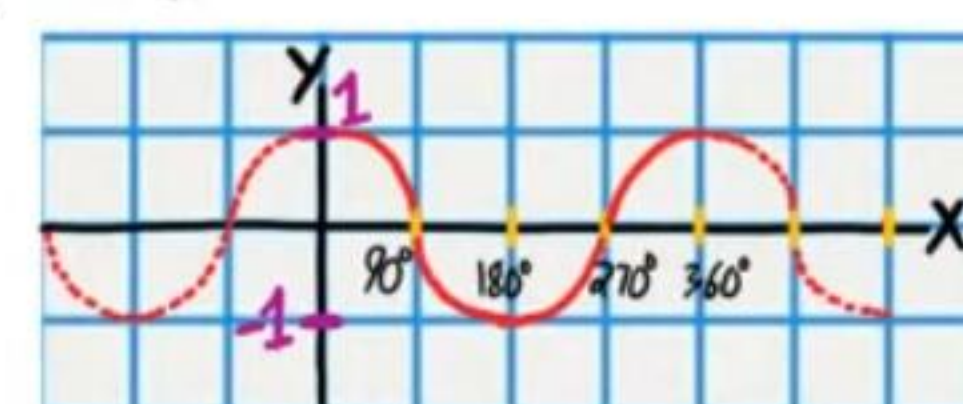


The Graphs of Basic Trigonometric Functions

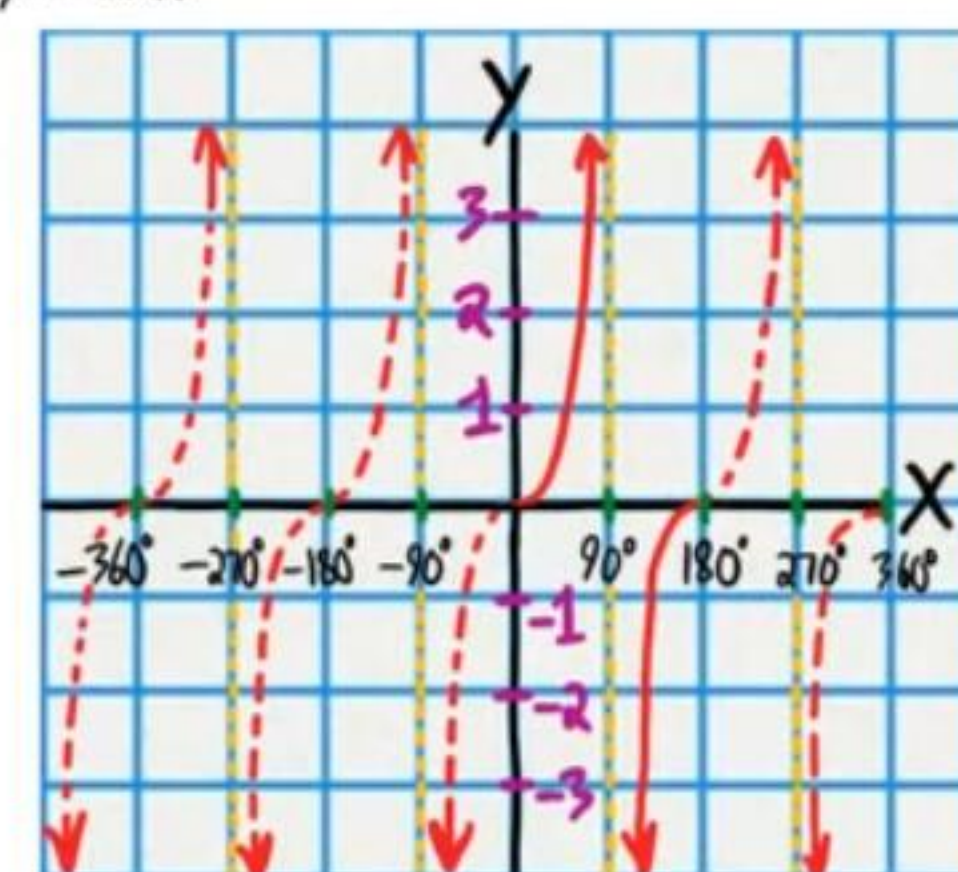
$$y = \sin x$$



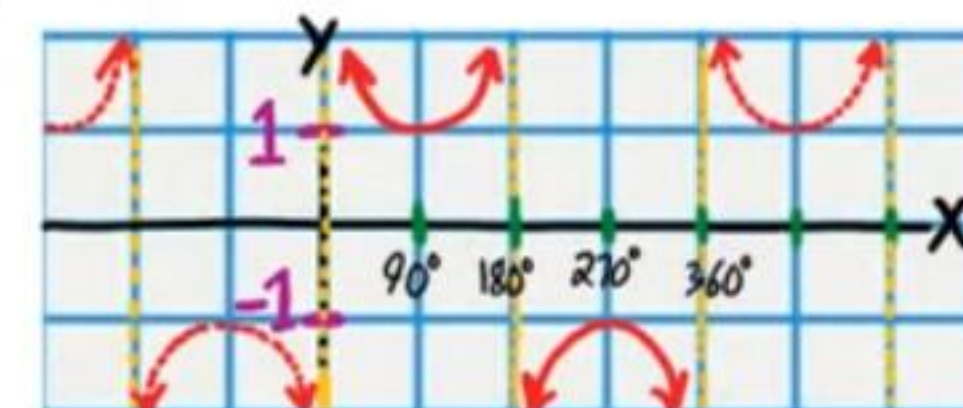
$$y = \cos x$$



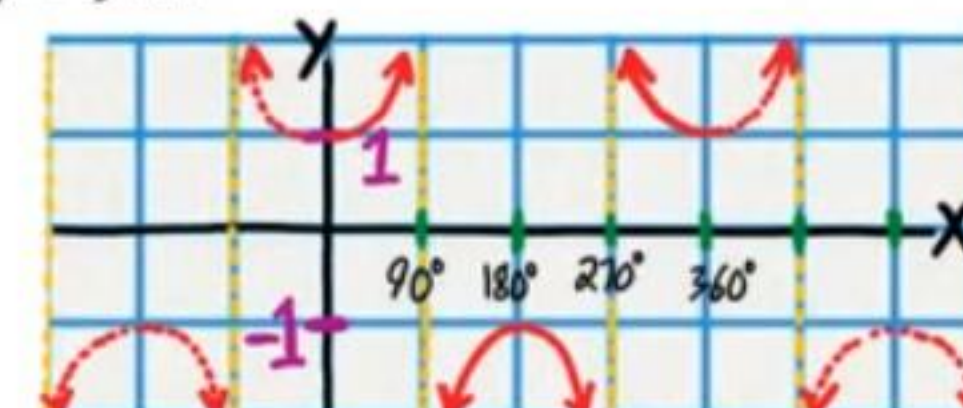
$$y = \tan x$$



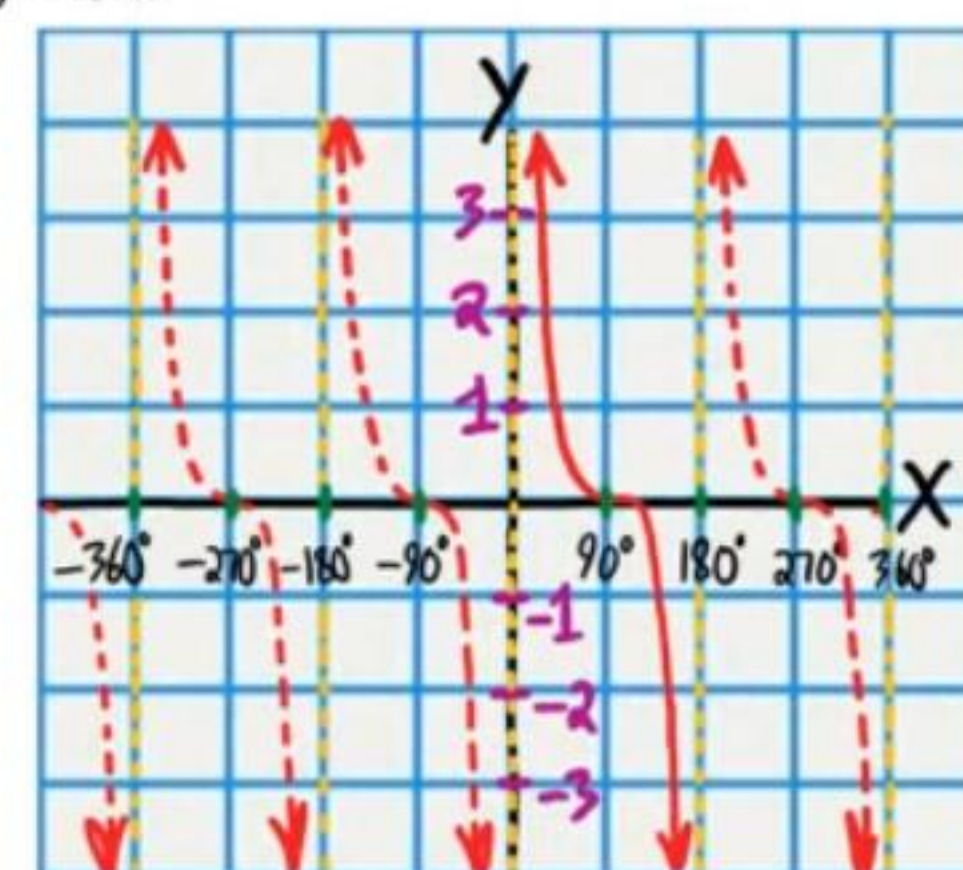
$$y = \csc x$$



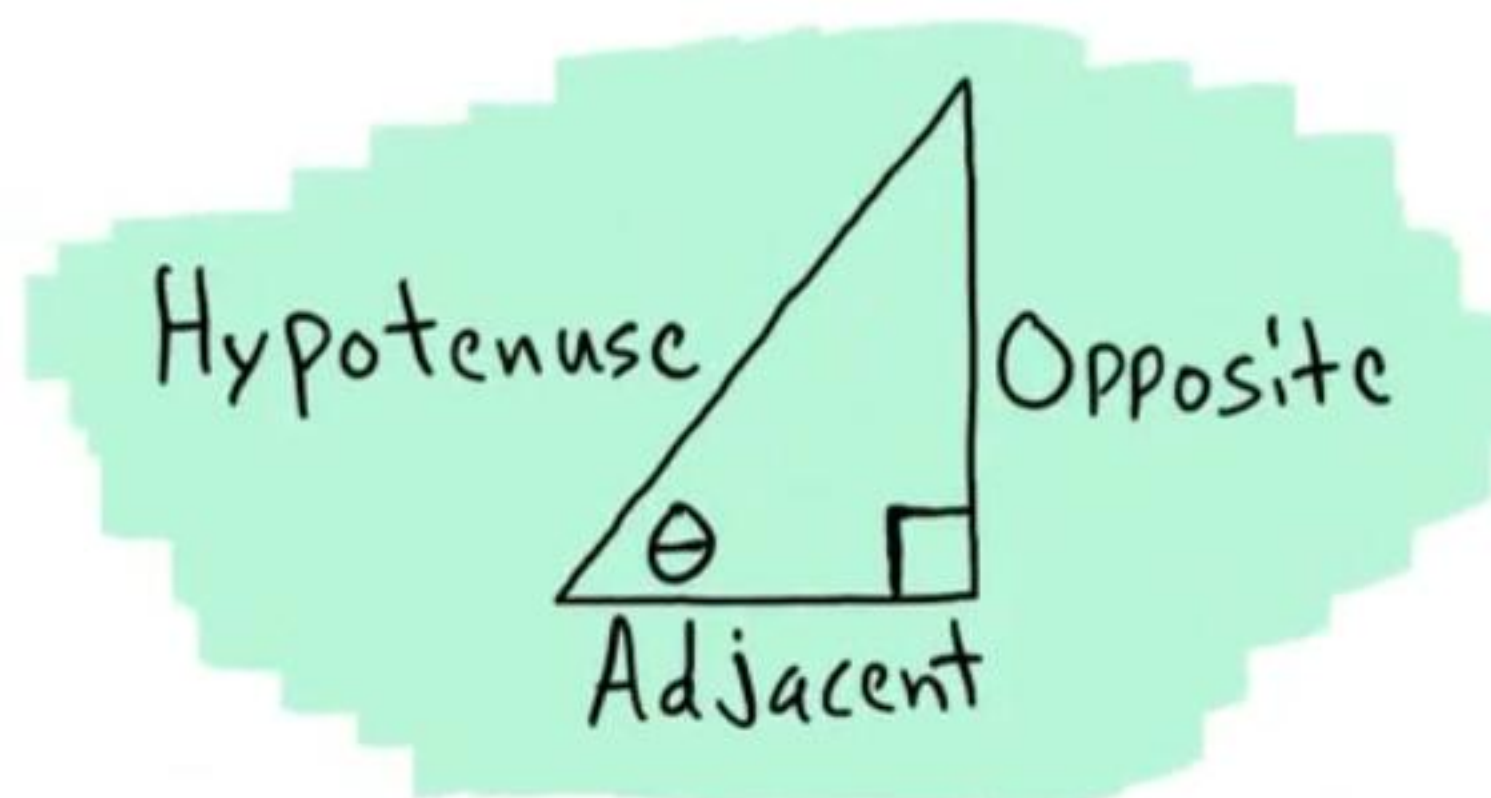
$$y = \sec x$$



$$y = \cot x$$



The Six Basic Trigonometric Functions



$$\sin \theta = \frac{\text{OPP}}{\text{HYP}}$$

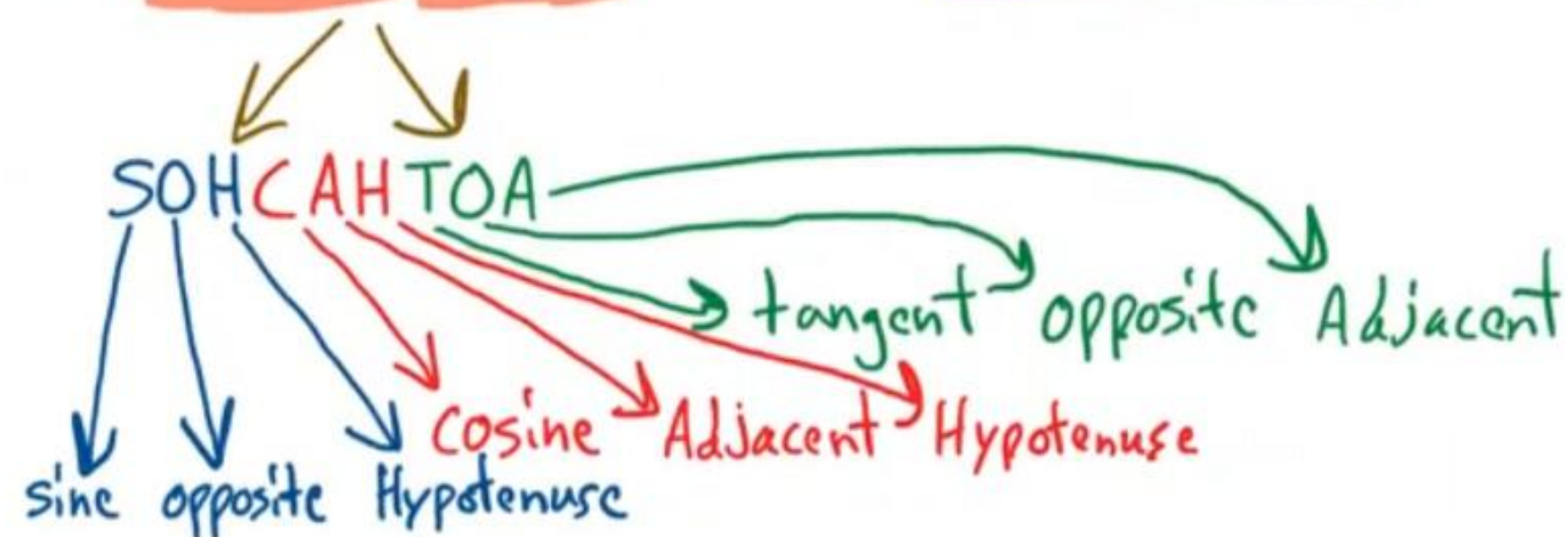
$$\csc \theta = \frac{\text{HYP}}{\text{OPP}}$$

$$\cos \theta = \frac{\text{Adj}}{\text{HYP}}$$

$$\sec \theta = \frac{\text{HYP}}{\text{Adj}}$$

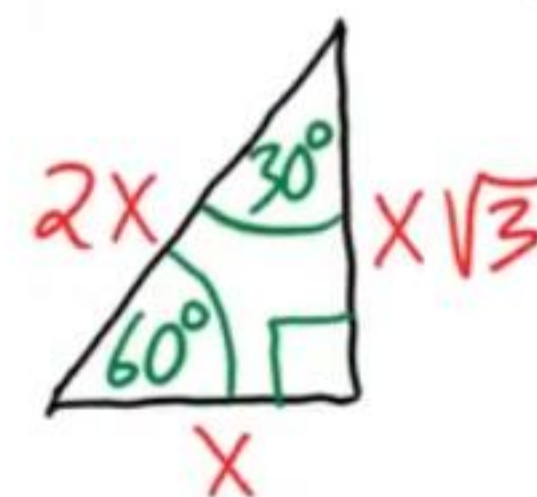
$$\tan \theta = \frac{\text{OPP}}{\text{Adj}}$$

$$\cot \theta = \frac{\text{Adj}}{\text{OPP}}$$

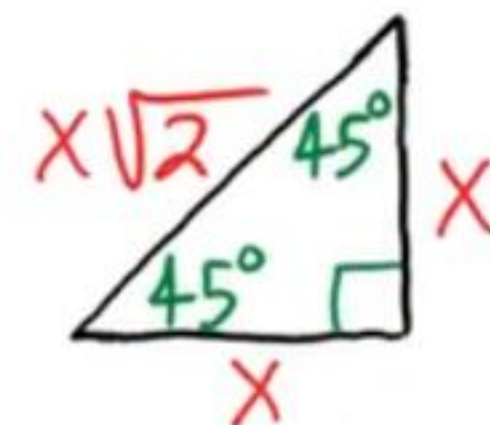


Special Right Triangles

30°-60°-90° Triangle Theorem: If the inner angles of a triangle are 30°, 60°, and 90°, and the length of the side opposite the 30° angle is X , then the length of the hypotenuse is $2X$ and the length of the side opposite the 60° angle is $X\sqrt{3}$.



45°-45°-90° Triangle Theorem: If the inner angles of a triangle are 45°, 45°, and 90°, and the length of the two sides opposite the 45° angles are X , then the length of the hypotenuse is $X\sqrt{2}$.



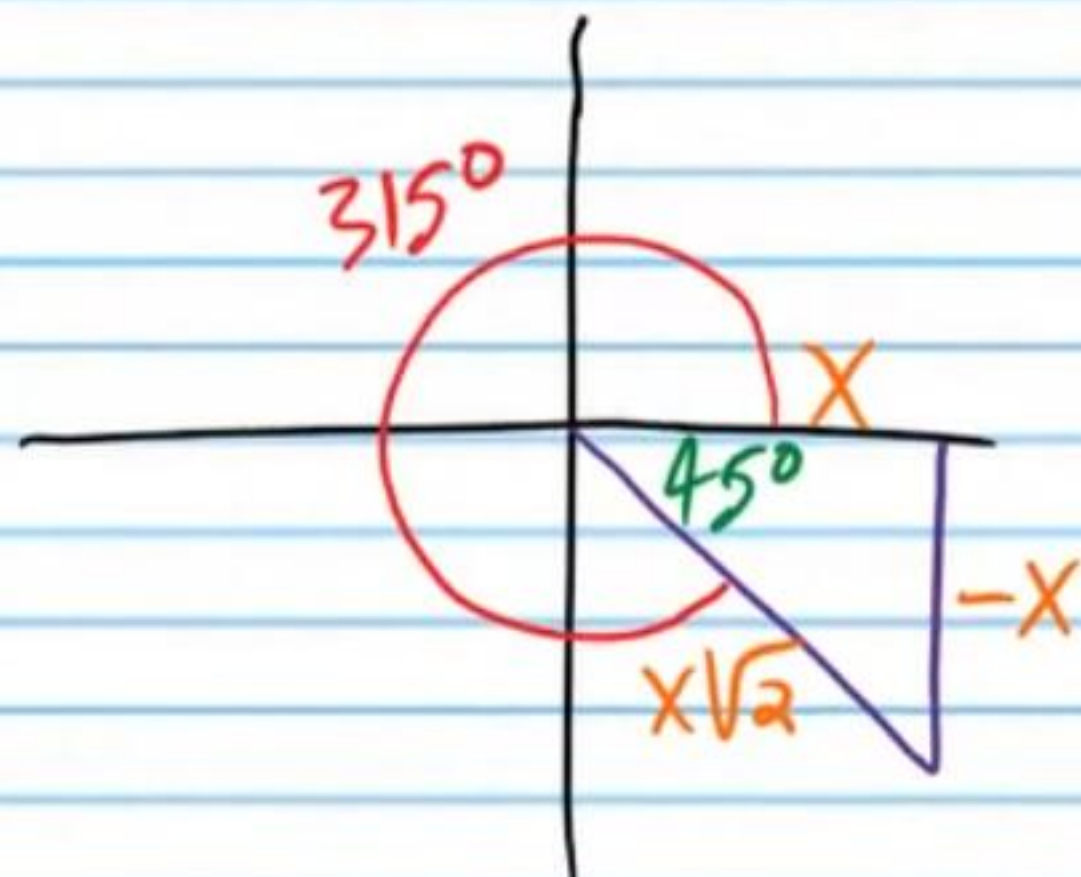
Find the value of each expression without a calculator.

• ① $\sec^{-1}(\csc 315^\circ)$

I) $\csc 315^\circ$

$$\frac{x\sqrt{2}}{-x}$$

$$-\sqrt{2}$$



II) $\sec^{-1}(\csc 315^\circ)$

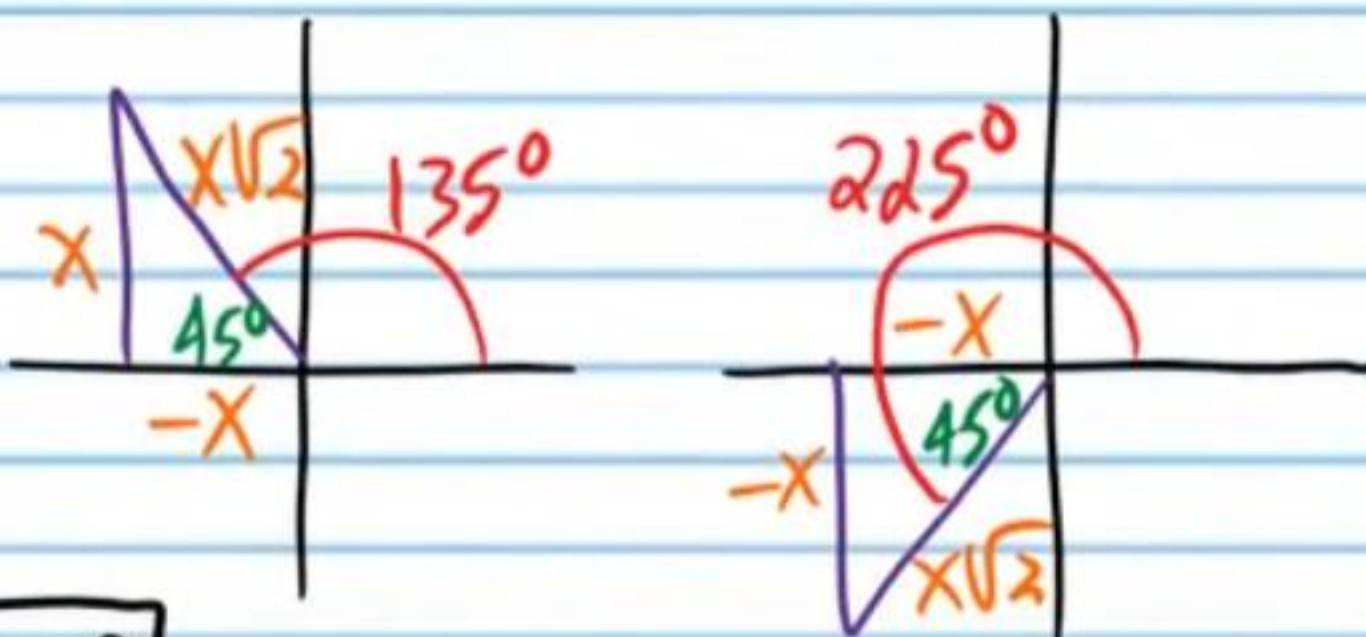
$$\sec^{-1}(-\sqrt{2})$$

$$\theta = \sec^{-1}(-\sqrt{2})$$

$$\sec \theta = -\sqrt{2}$$

$$\theta = 135^\circ, 225^\circ$$

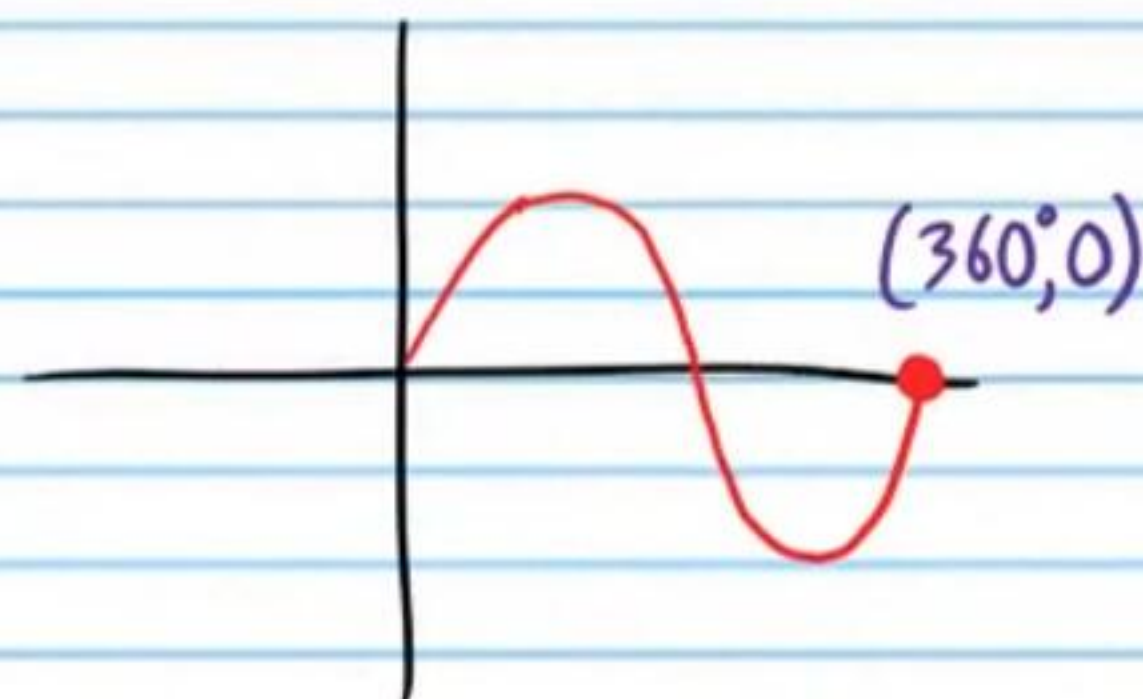
$$\sec^{-1}(\csc 315^\circ) = \boxed{135^\circ}$$



② $\cos^{-1}(\sin 360^\circ)$

I) $\sin 360^\circ$

$$\boxed{0}$$



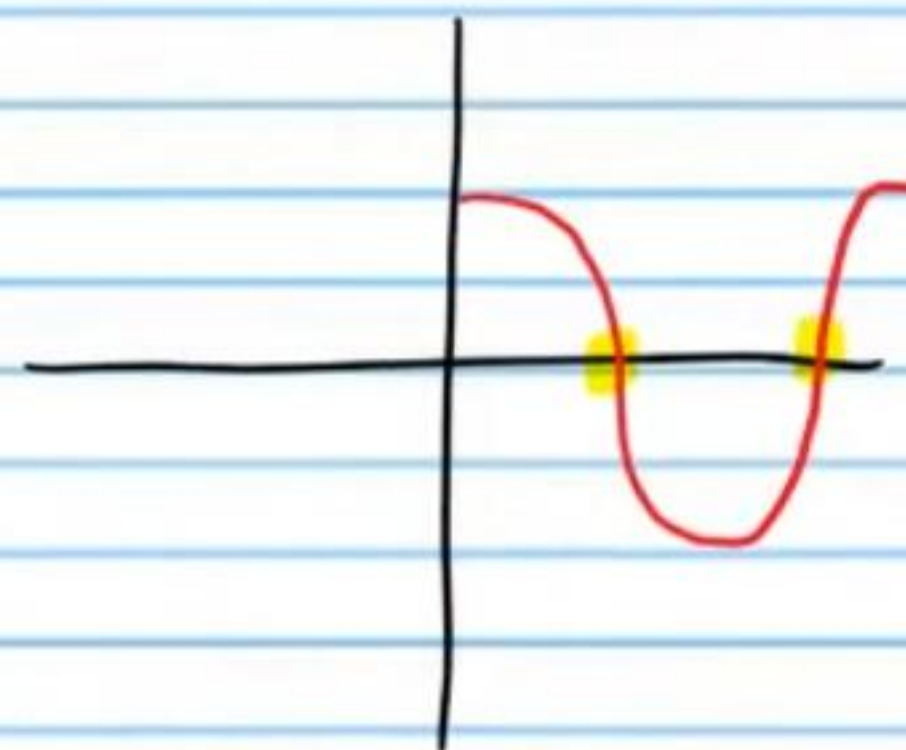
II) $\cos^{-1}(\sin 360^\circ)$

$$\cos^{-1}(0)$$

$$\theta = \cos^{-1}(0)$$

$$\cos \theta = 0$$

$$\theta = 90^\circ, 270^\circ$$



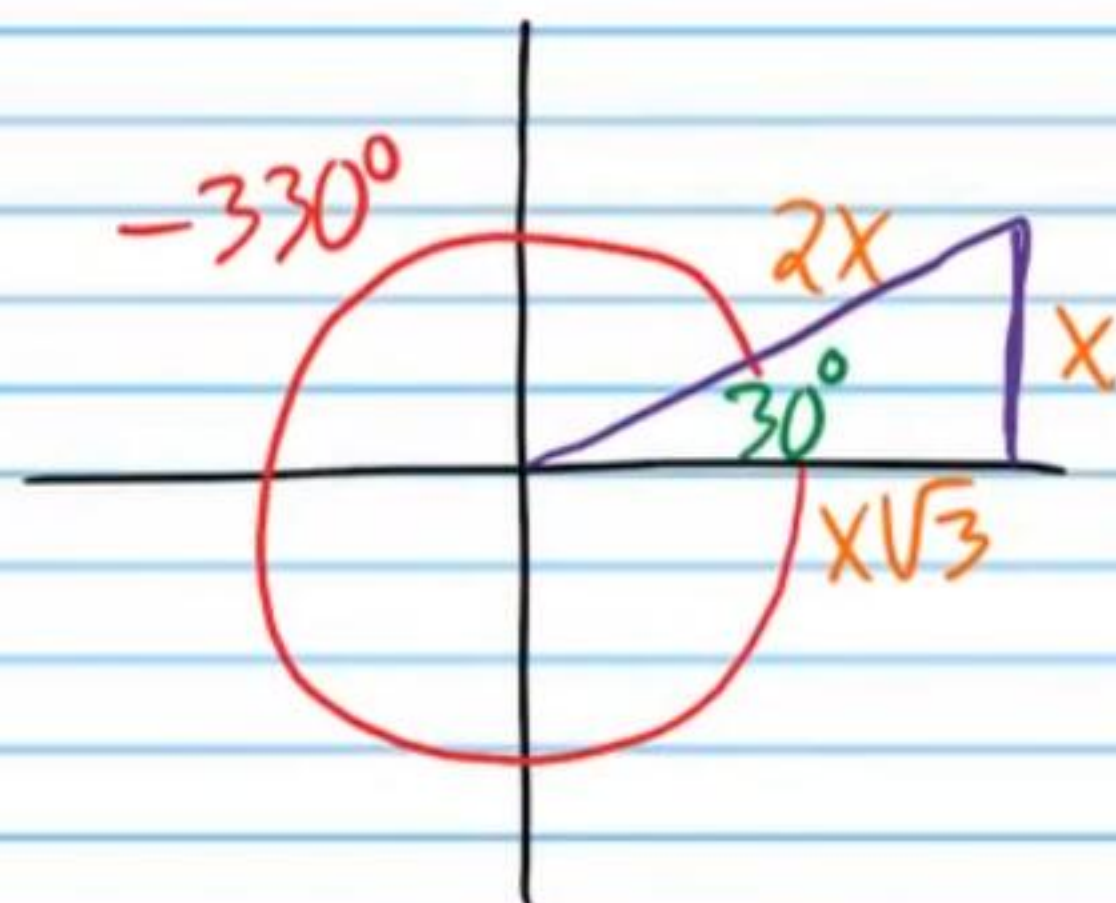
$$\cos^{-1}(\sin 360^\circ) = \boxed{90^\circ}$$

$$\textcircled{3} \cot^{-1}(\tan(-330^\circ))$$

$$\text{I) } \tan(-330^\circ)$$

$$\frac{x}{x\sqrt{3}}$$

$$\frac{1}{\sqrt{3}}$$



$$\text{II) } \cot^{-1}(\tan(-330^\circ))$$

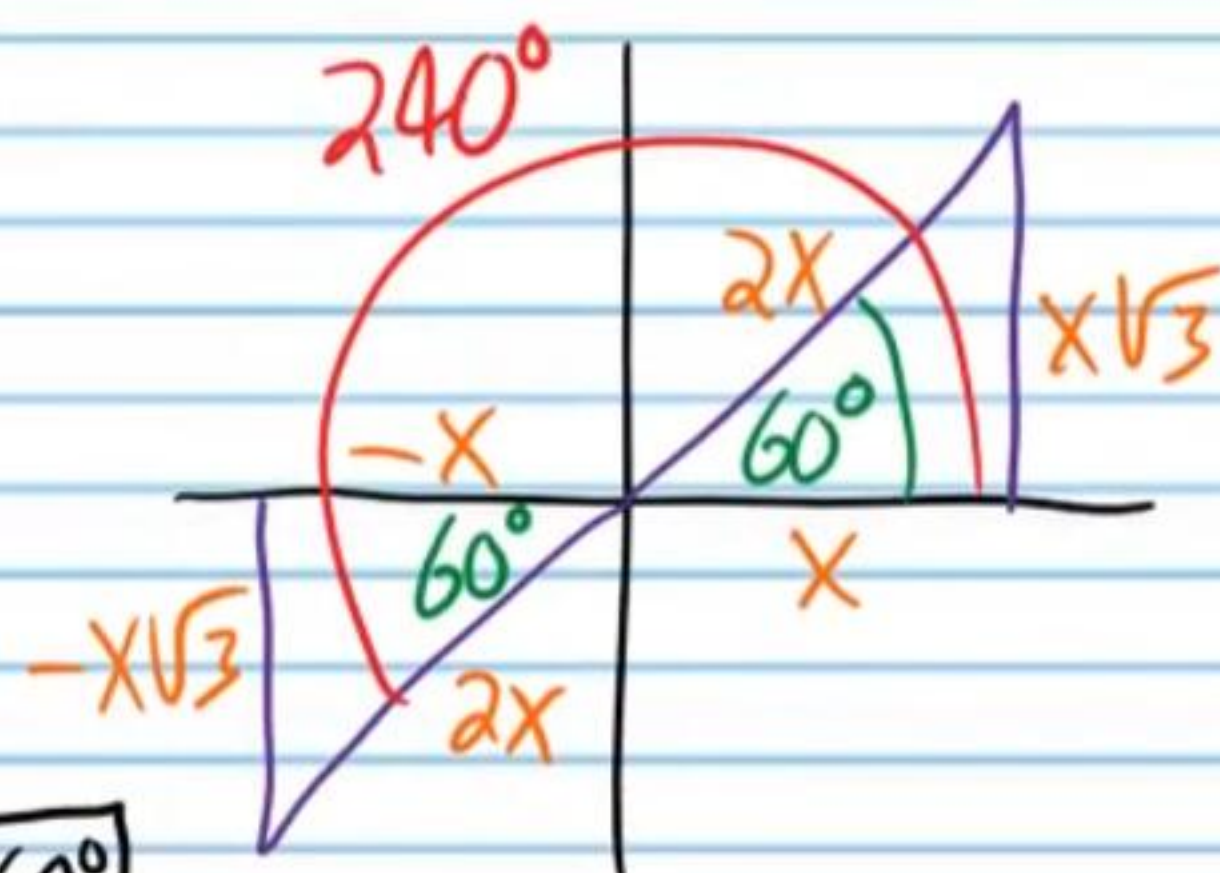
$$\cot^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$\theta = \cot^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$\cot \theta = \frac{1}{\sqrt{3}}$$

$$\theta = 60^\circ, 240^\circ$$

$$\cot^{-1}(\tan(-330^\circ)) = \boxed{60^\circ}$$

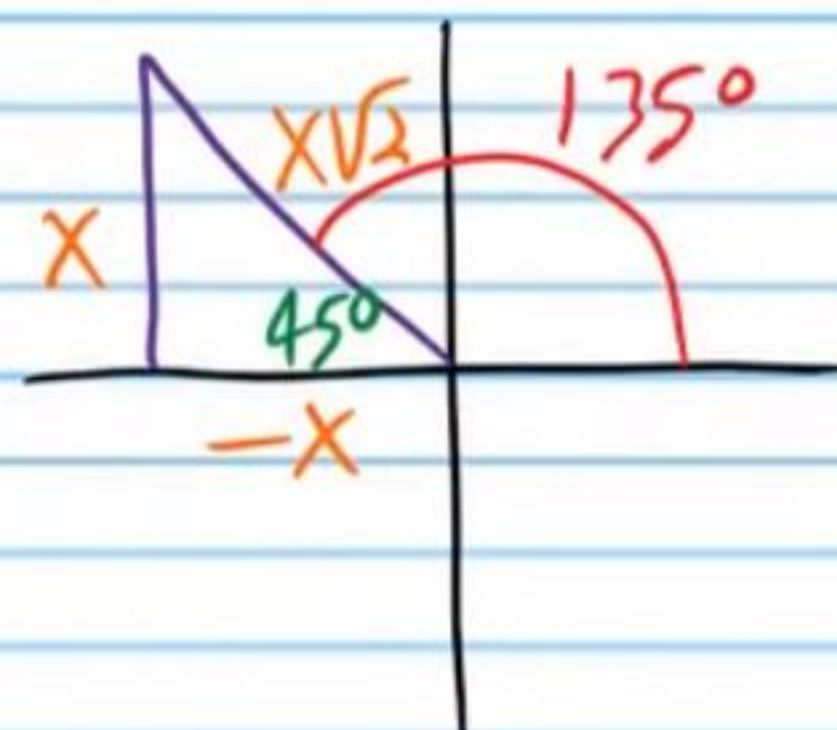


$$\textcircled{4} \sin^{-1}(\cos 135^\circ)$$

$$\text{I) } \cos 135^\circ$$

$$\frac{-x}{x\sqrt{2}}$$

$$-\frac{1}{\sqrt{2}}$$



$$\text{II) } \sin^{-1}(\cos 135^\circ)$$

$$\sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)$$

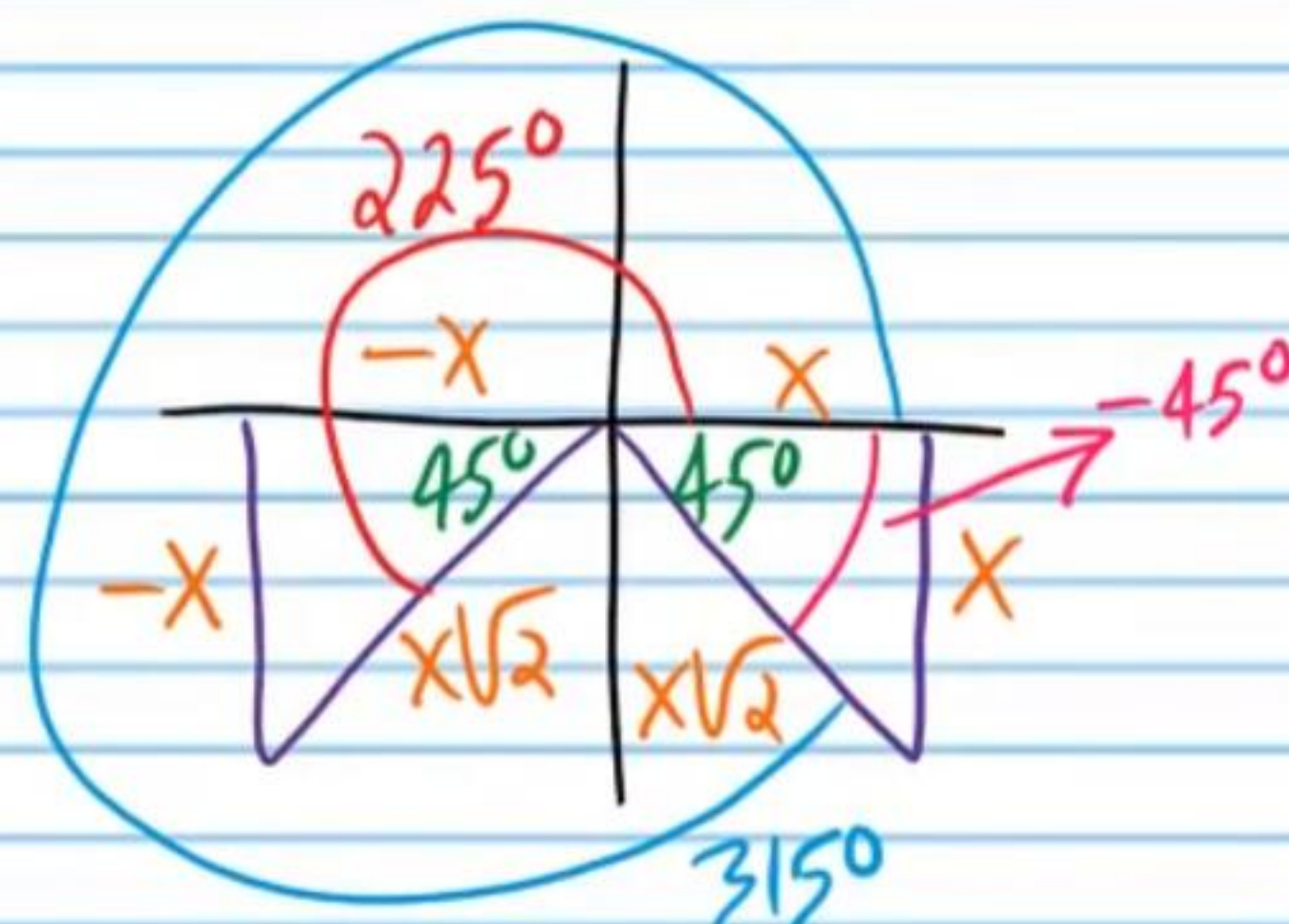
$$\theta = \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)$$

$$\sin \theta = -\frac{1}{\sqrt{2}}$$

$$\theta = 225^\circ, 315^\circ$$

$$\theta = -45^\circ$$

$$\sin^{-1}(\cos 135^\circ) = \boxed{-45^\circ}$$

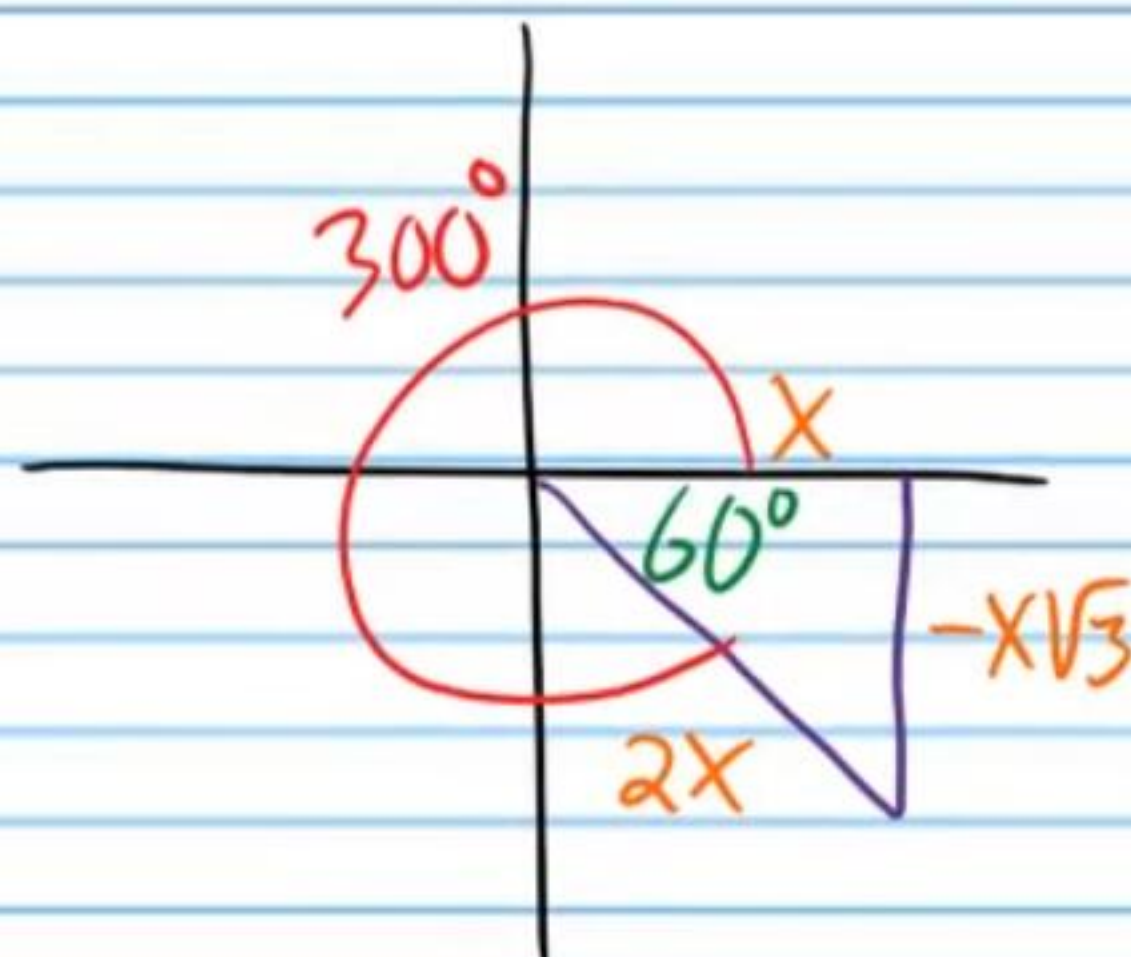


• ⑤ $\csc^{-1}(\sec 300^\circ)$

I) $\sec 300^\circ$

$$\frac{2x}{x}$$

$$2$$



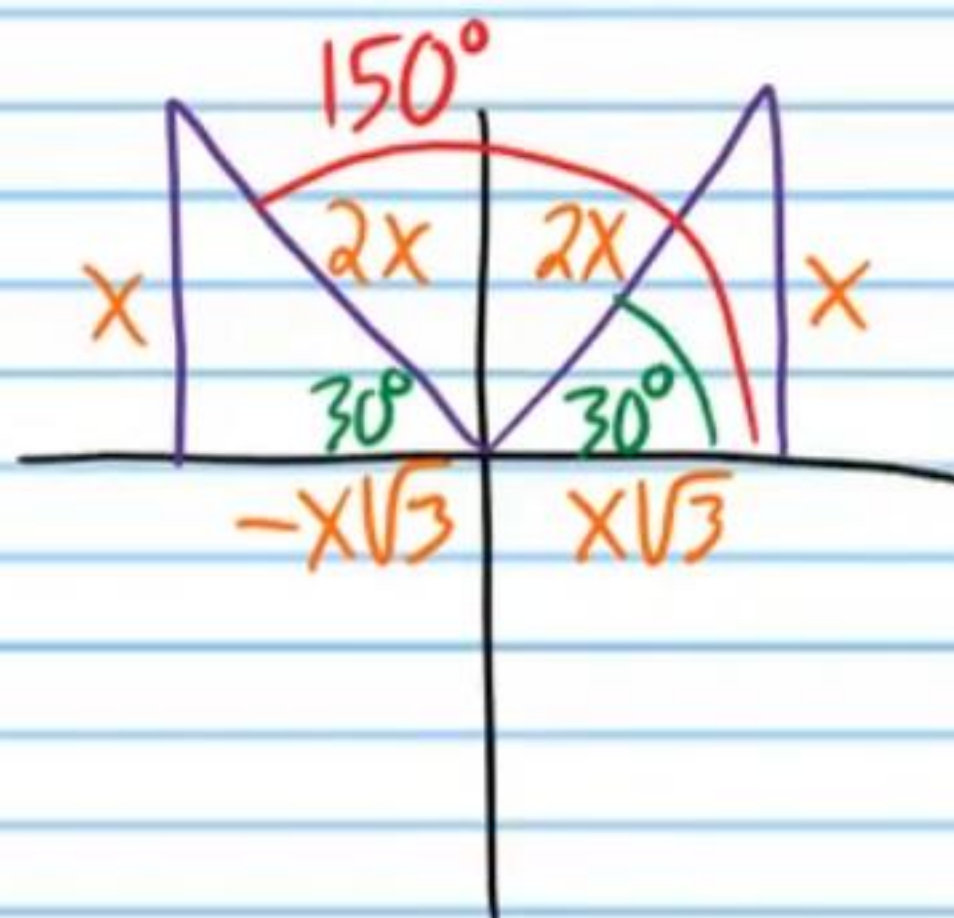
II) $\csc^{-1}(\sec 300^\circ)$

$$\csc^{-1} 2$$

$$\theta = \csc^{-1} 2$$

$$\csc \theta = 2$$

$$\theta = 30^\circ, 150^\circ$$



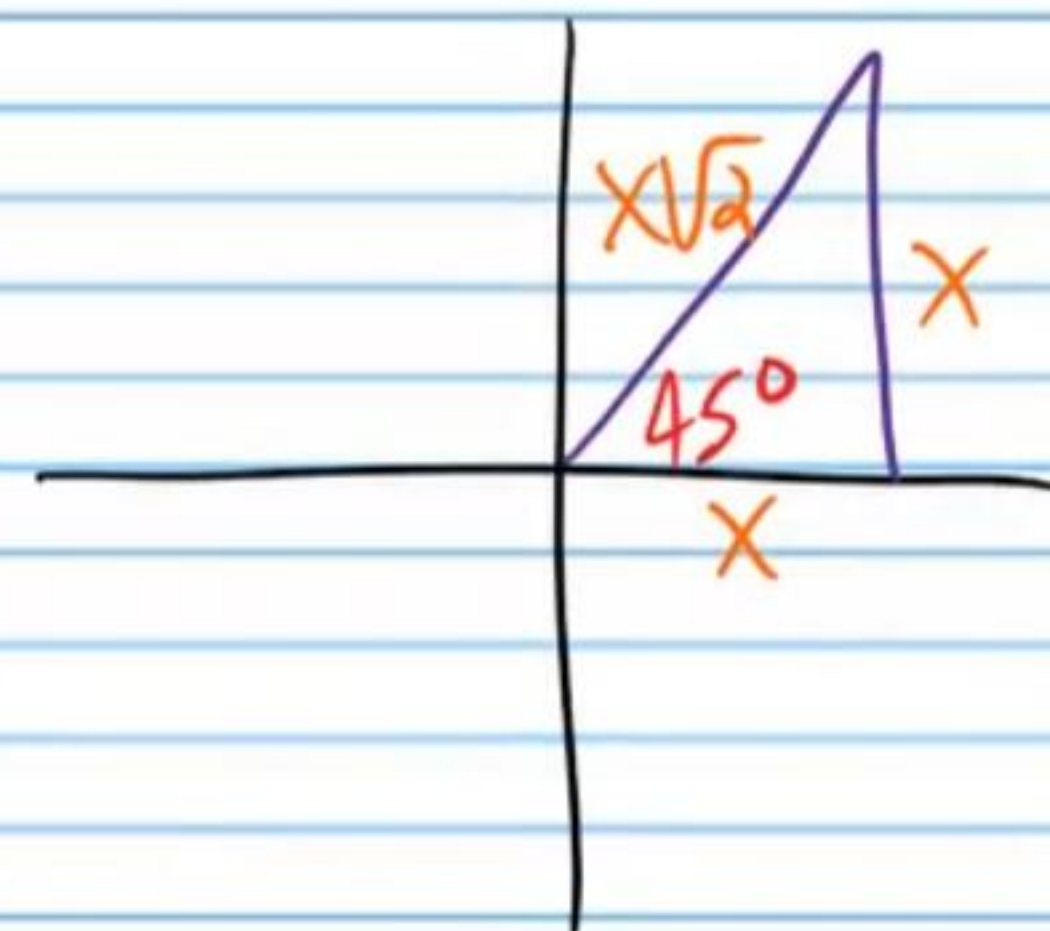
$$\csc^{-1}(\sec 300^\circ) = \boxed{30^\circ}$$

⑥ $\cos^{-1}(\sin 45^\circ)$

I) $\sin 45^\circ$

$$\frac{x}{x\sqrt{2}}$$

$$\frac{1}{\sqrt{2}}$$



II) $\cos^{-1}(\sin 45^\circ)$

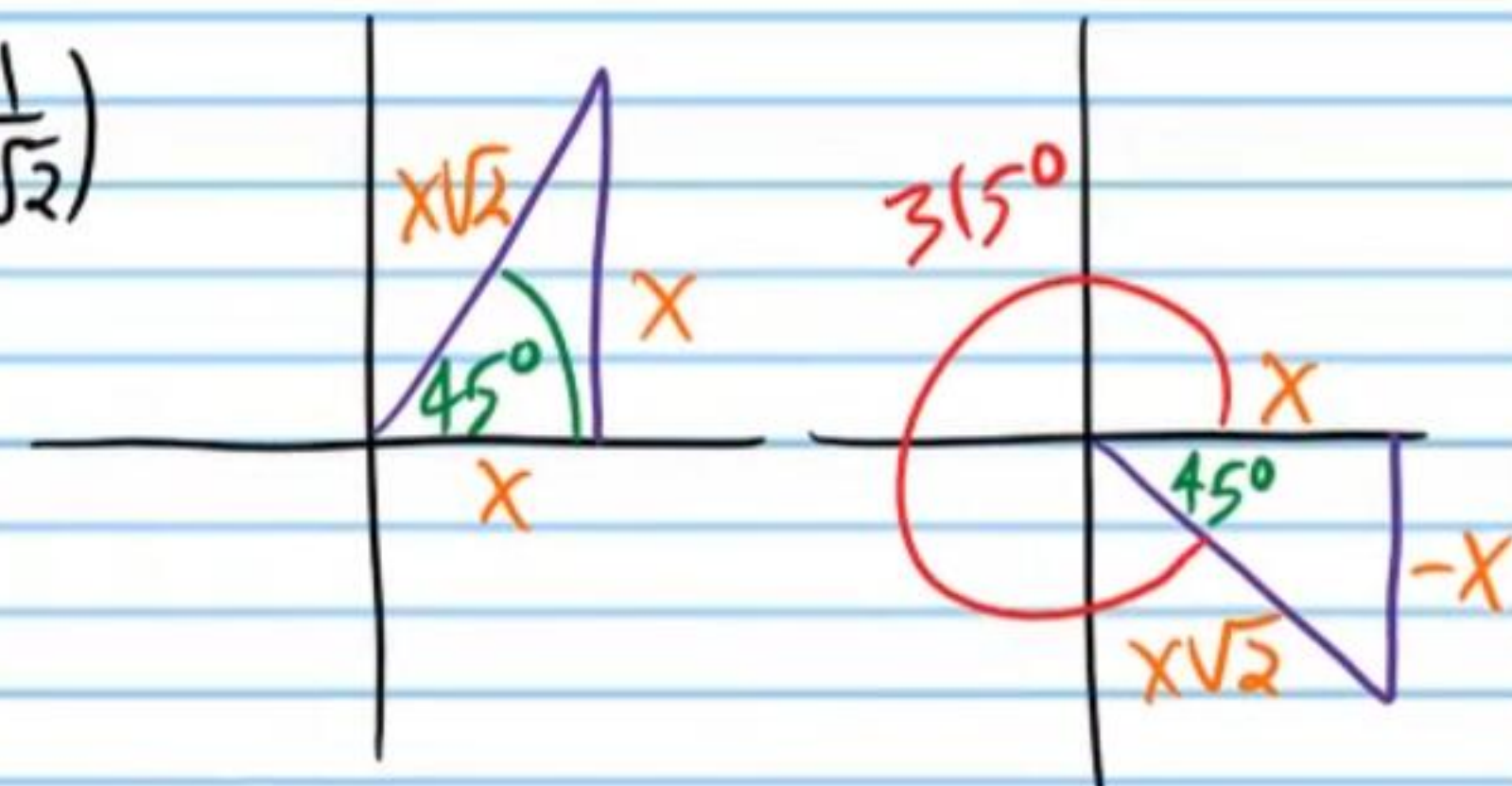
$$\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$\cos \theta = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ, 315^\circ$$

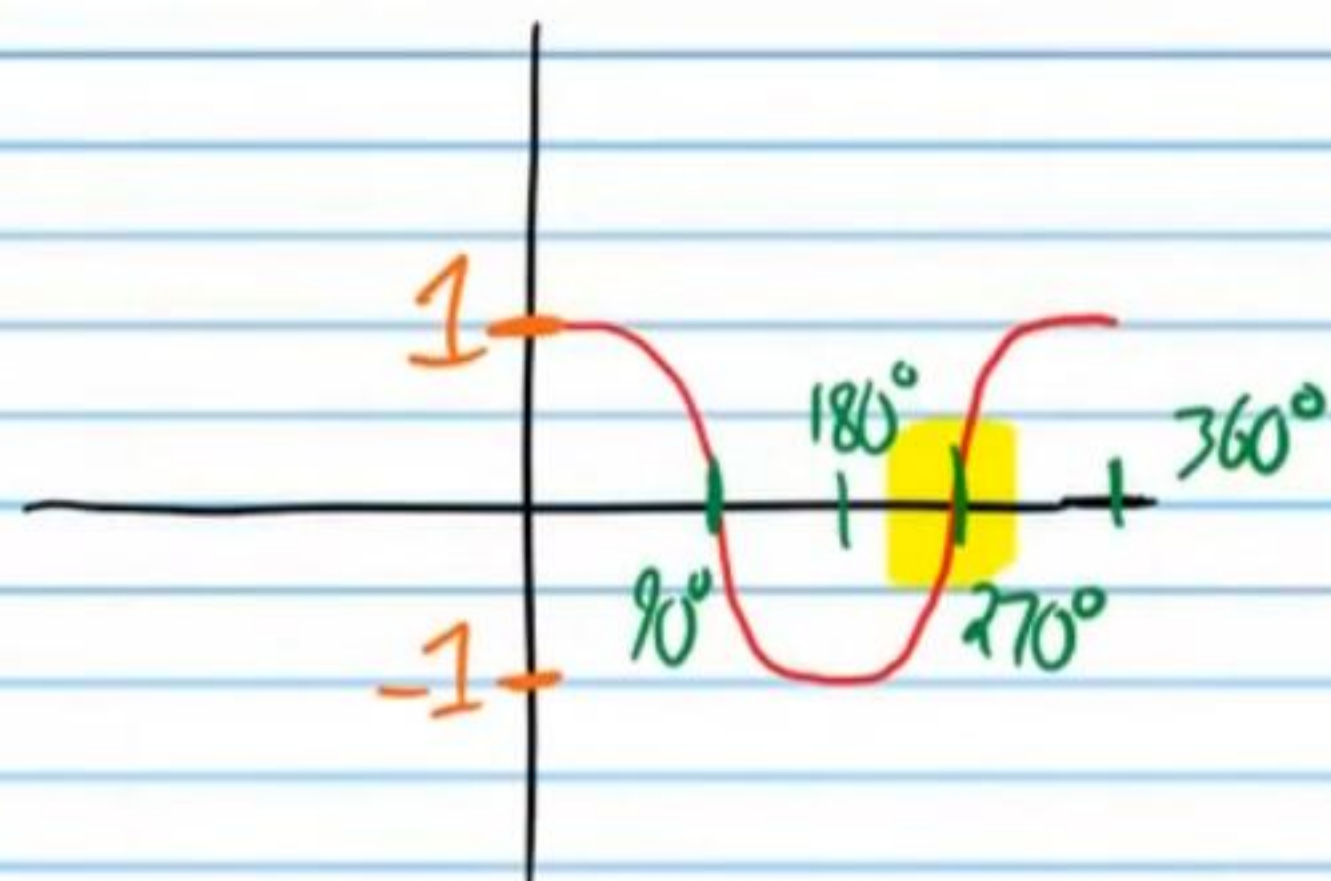
$$\cos^{-1}(\sin 45^\circ) = \boxed{45^\circ}$$



⑦ $\sin^{-1}(\cos 270^\circ)$

I) $\cos 270^\circ$

0



II) $\sin^{-1}(\cos 270^\circ)$

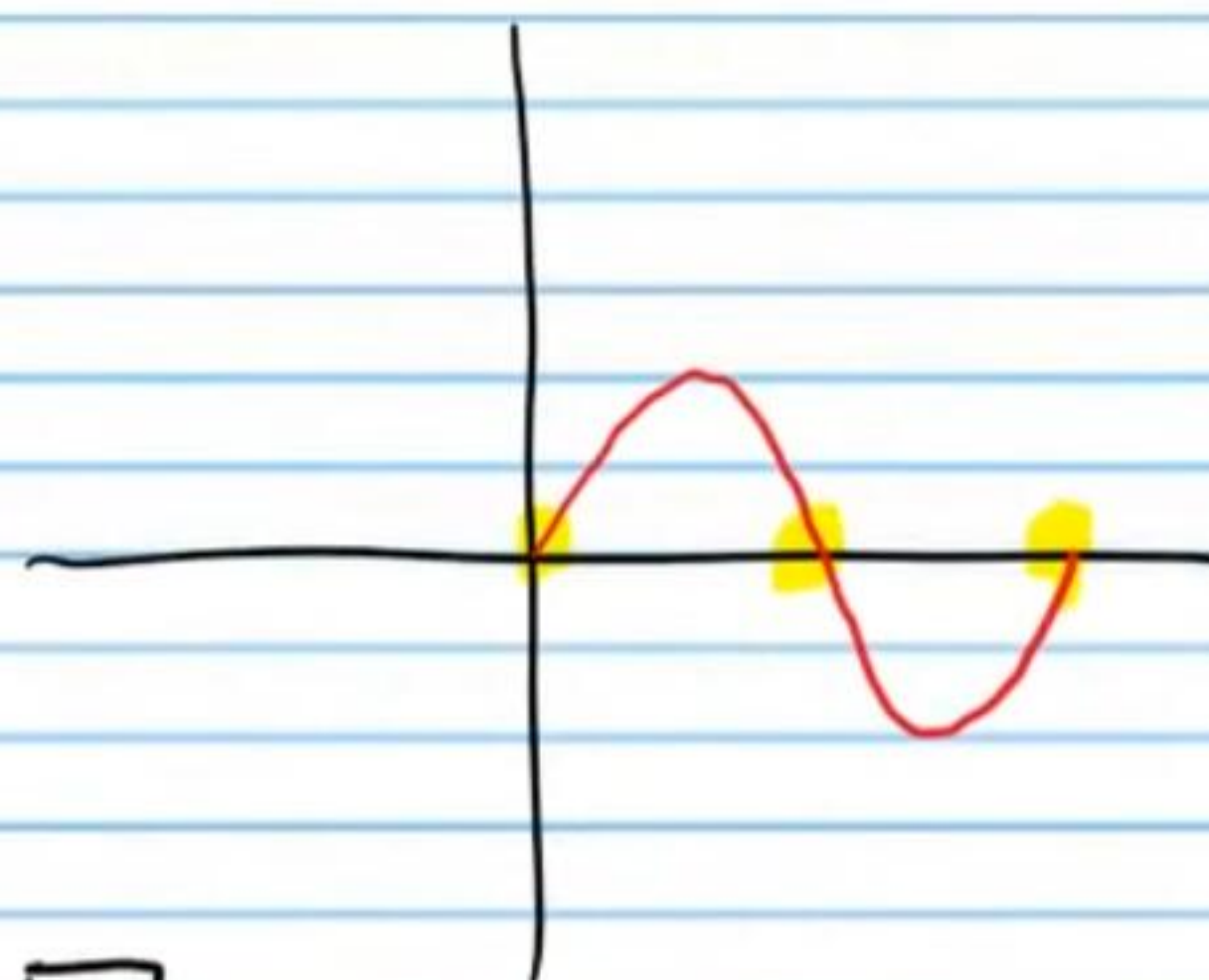
$\sin^{-1}(0)$

$\theta = \sin^{-1}(0)$

$\sin \theta = 0$

$\theta = 0^\circ, 180^\circ, 360^\circ$

$\sin^{-1}(\cos 270^\circ) = \boxed{0^\circ}$



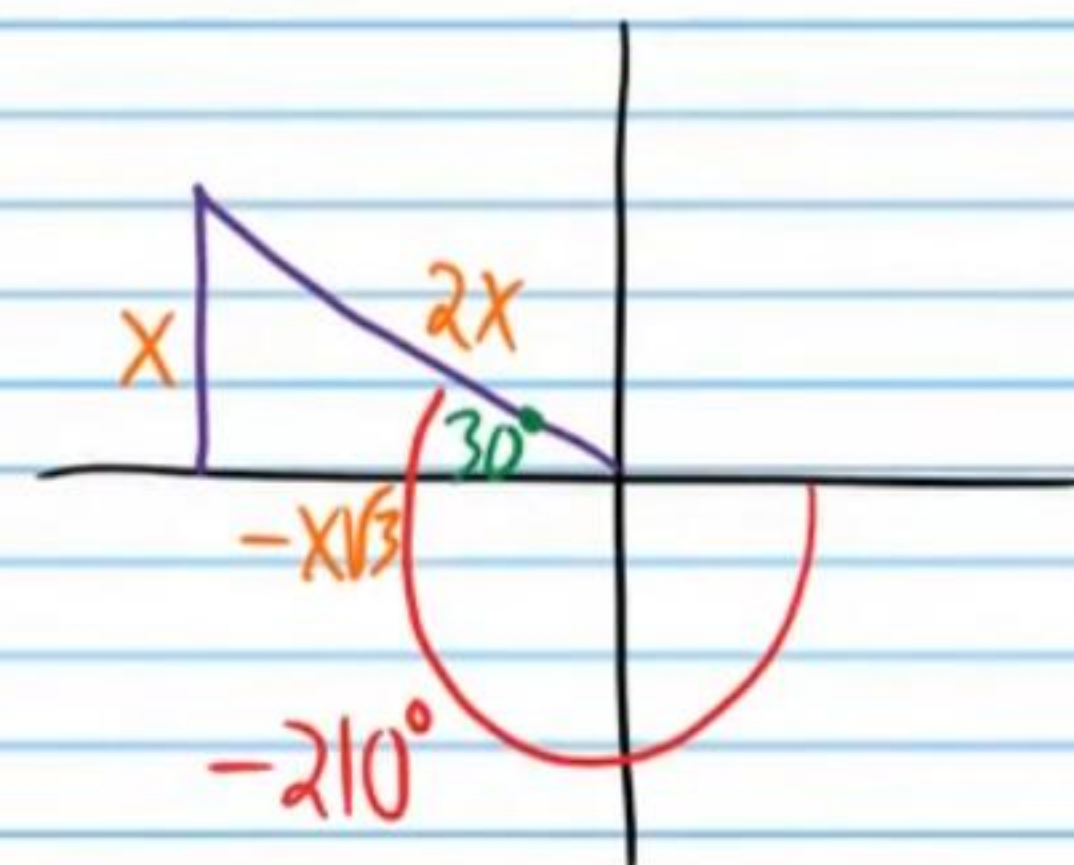
Your answer will be negative.

⑧ $\tan^{-1}(\cot(-210^\circ))$

I) $\cot(-210^\circ)$

$\frac{-X\sqrt{3}}{X}$

$-\sqrt{3}$



II) $\tan^{-1}(\cot(-210^\circ))$

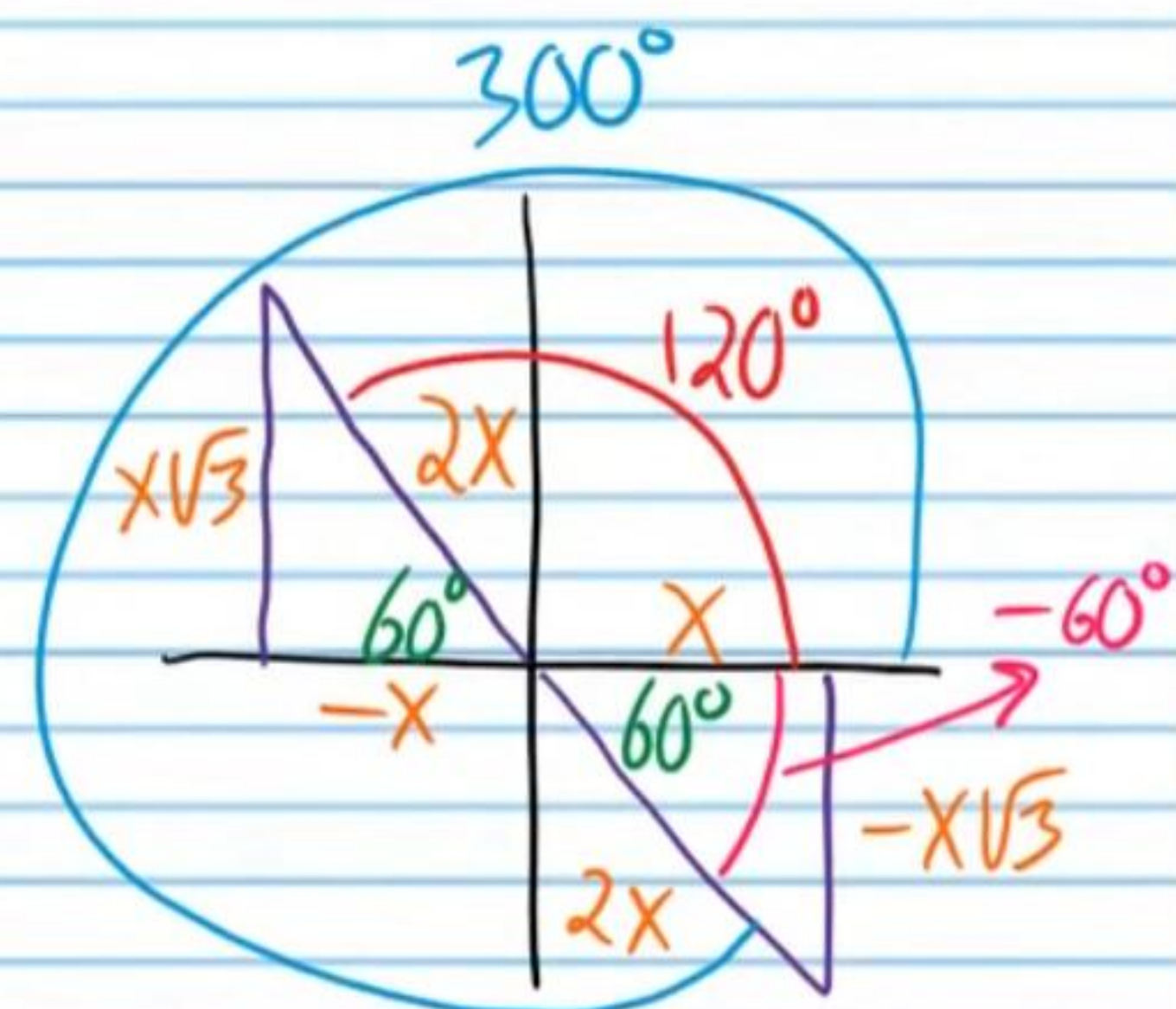
$\tan^{-1}(-\sqrt{3})$

$\theta = \tan^{-1}(-\sqrt{3})$

$\tan \theta = -\sqrt{3}$

$\theta = 120^\circ, 300^\circ$

$\theta = -60^\circ$



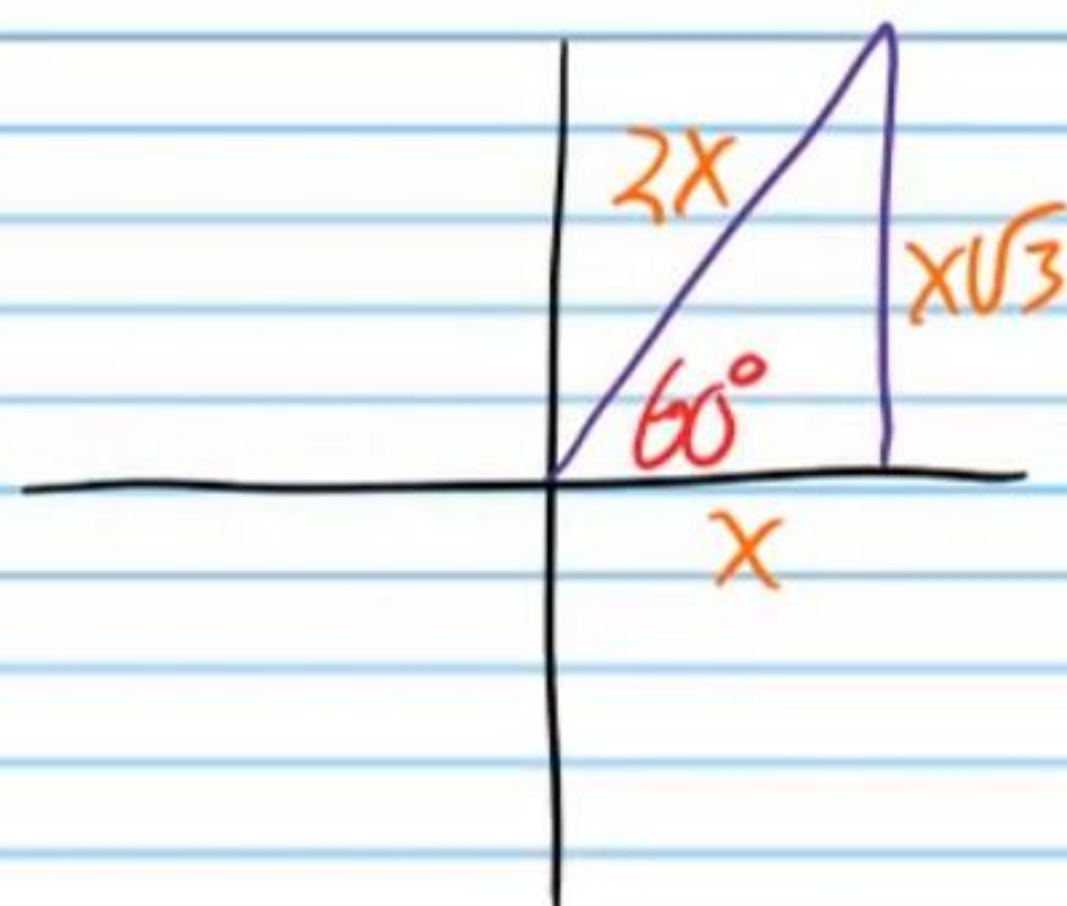
$\tan^{-1}(\cot(-210^\circ)) = \boxed{-60^\circ}$

• ⑨ $\csc^{-1}(\csc 60^\circ)$

I) $\csc 60^\circ$

$$\frac{2x}{x\sqrt{3}}$$

$$\frac{2}{\sqrt{3}}$$



II) $\csc^{-1}(\csc 60^\circ)$

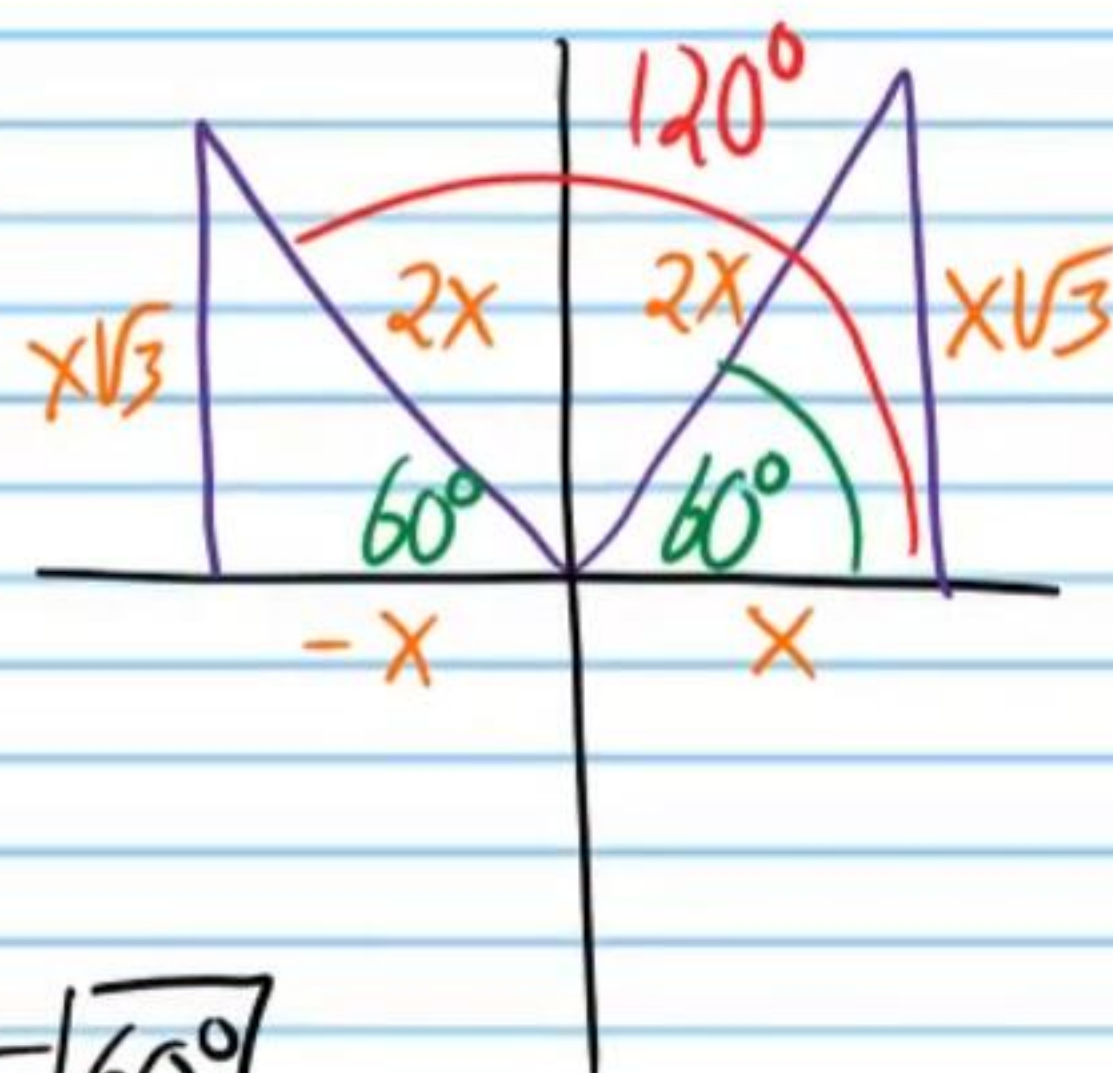
$$\csc^{-1}\left(\frac{2}{\sqrt{3}}\right)$$

$$\theta = \csc^{-1}\left(\frac{2}{\sqrt{3}}\right)$$

$$\csc \theta = \frac{2}{\sqrt{3}}$$

$$\theta = 60^\circ, 120^\circ$$

$$\csc^{-1}(\csc 60^\circ) = \boxed{60^\circ}$$



⑩ $\cos^{-1}(\cos 90^\circ)$

$$\boxed{90^\circ}$$

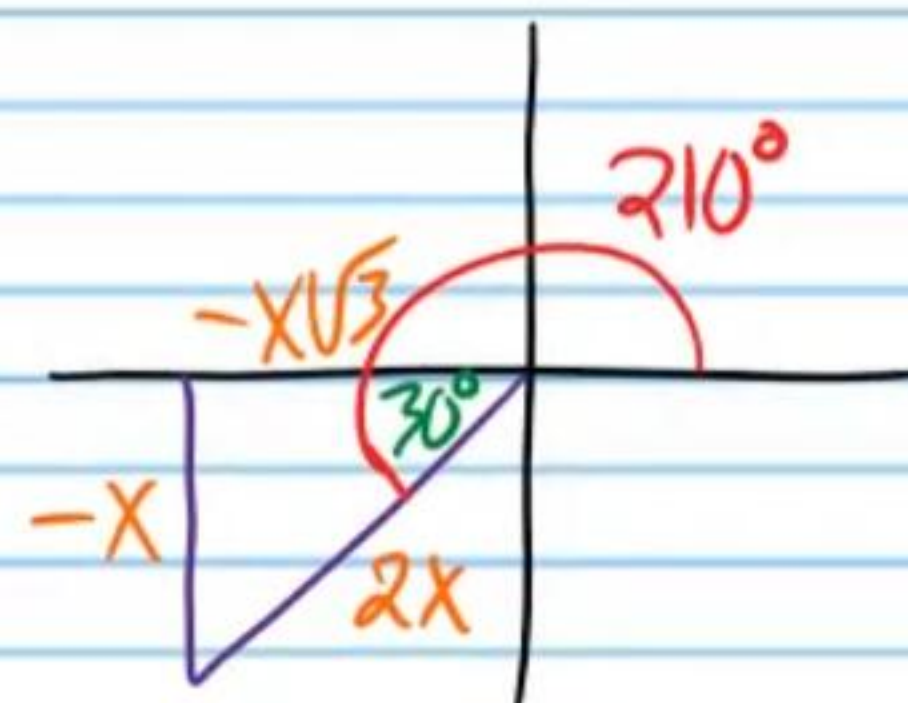
⑪ $\sin^{-1}(\sin 210^\circ)$

$$\boxed{120^\circ}$$

I) $\sin 210^\circ$

$$\frac{-x}{2x}$$

$$-\frac{1}{2}$$



II) $\sin^{-1}(\sin 210^\circ)$

$$\sin^{-1}\left(-\frac{1}{2}\right)$$

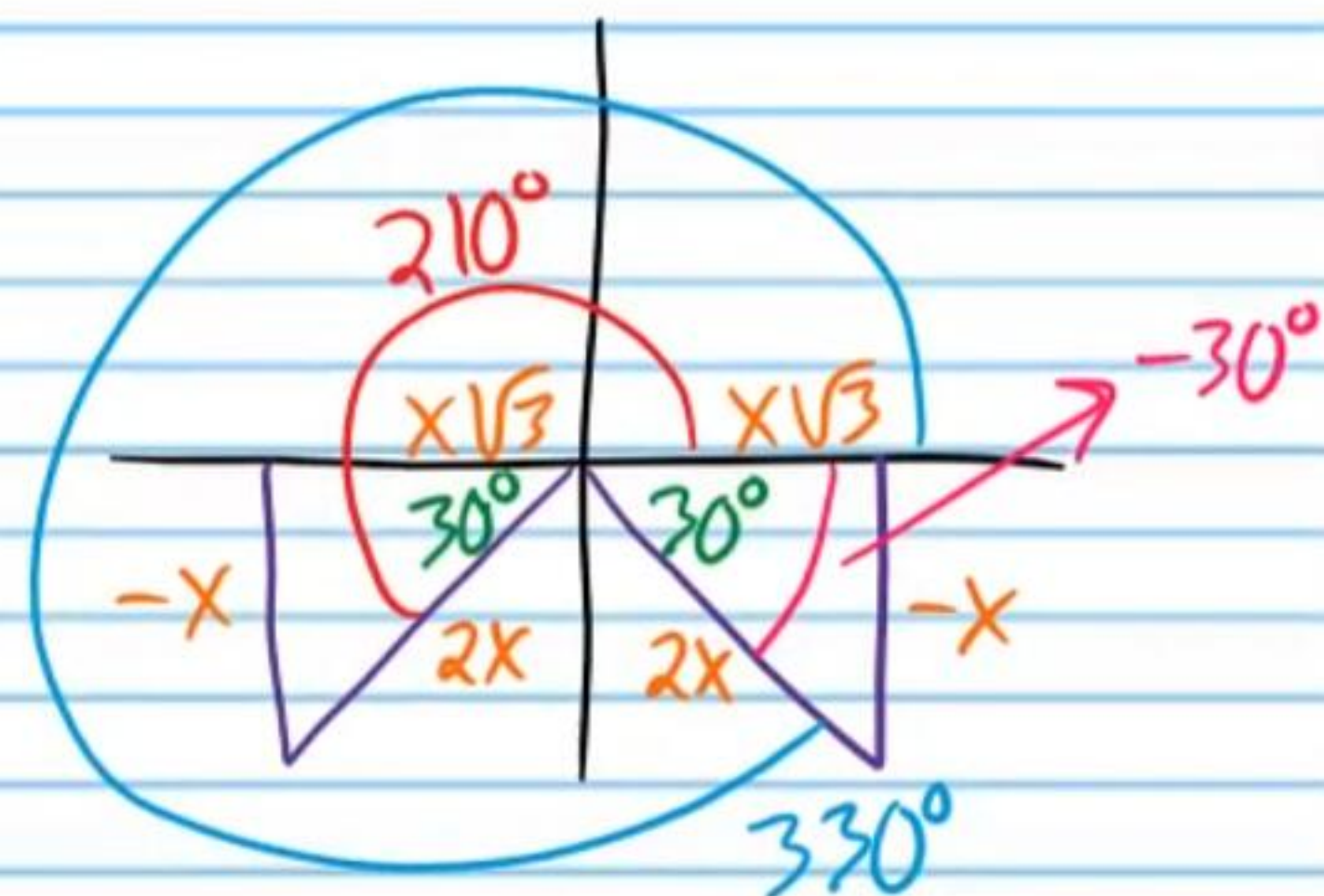
$$\theta = \sin^{-1}\left(-\frac{1}{2}\right)$$

$$\sin \theta = -\frac{1}{2}$$

$$\theta = 210^\circ, 330^\circ$$

$$\theta = -30^\circ$$

$$\sin^{-1}(\sin 210^\circ) = \boxed{-30^\circ}$$

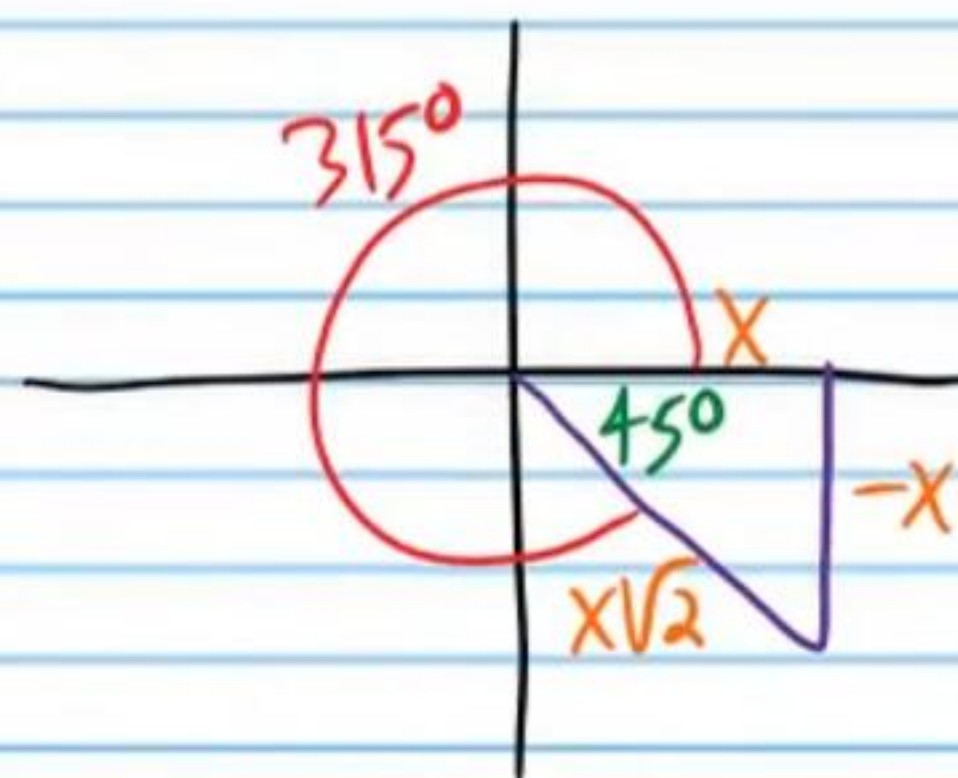


⑬ $\sec^{-1}(\sec 315^\circ)$

I) $\sec 315^\circ$

$$\frac{x\sqrt{2}}{x}$$

$$\sqrt{2}$$



II) $\sec^{-1}(\sec 315^\circ)$

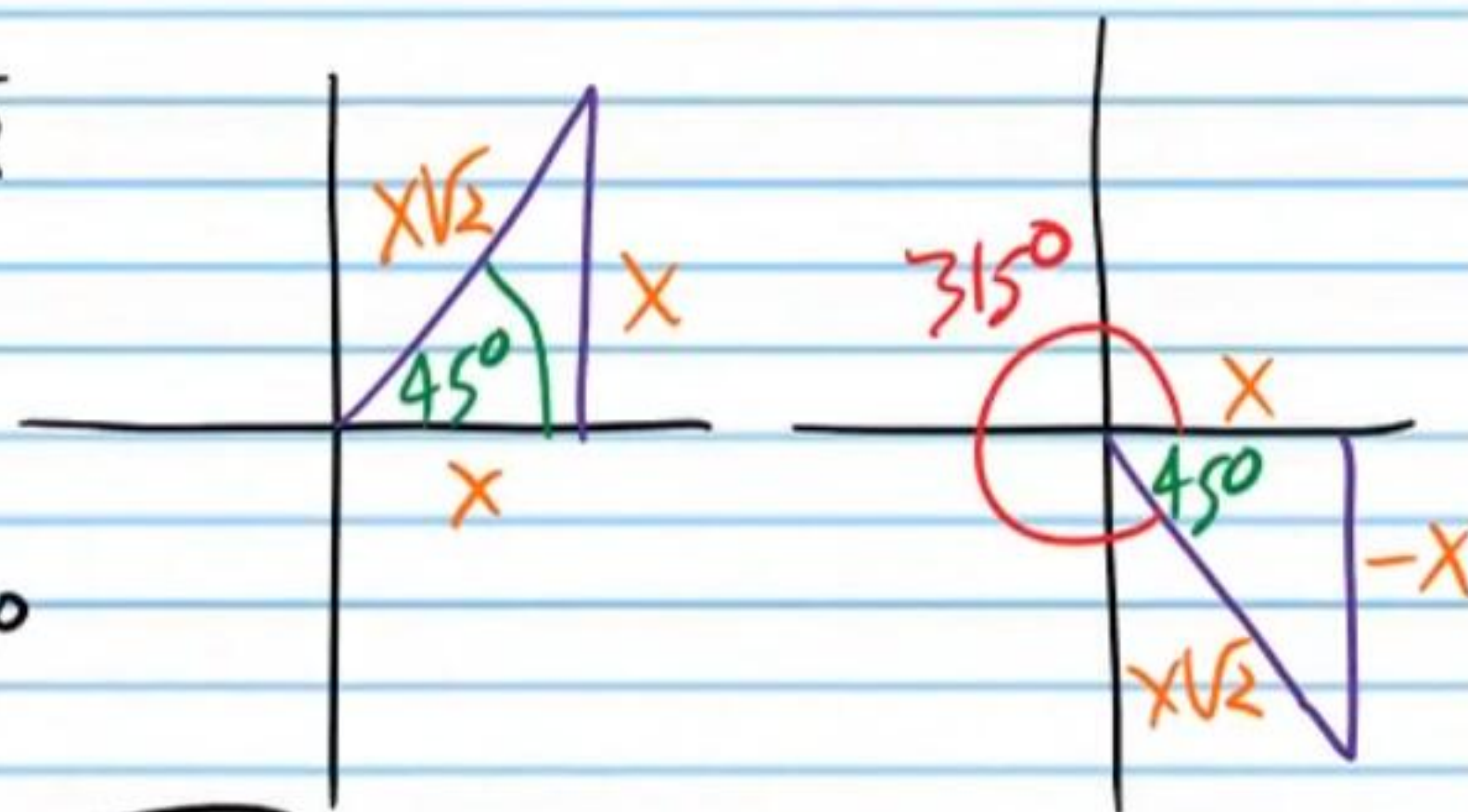
$$\sec^{-1} \sqrt{2}$$

$$\theta = \sec^{-1} \sqrt{2}$$

$$\sec \theta = \sqrt{2}$$

$$\theta = 45^\circ, 315^\circ$$

$$\sec^{-1}(\sec 315^\circ) = \boxed{45^\circ}$$

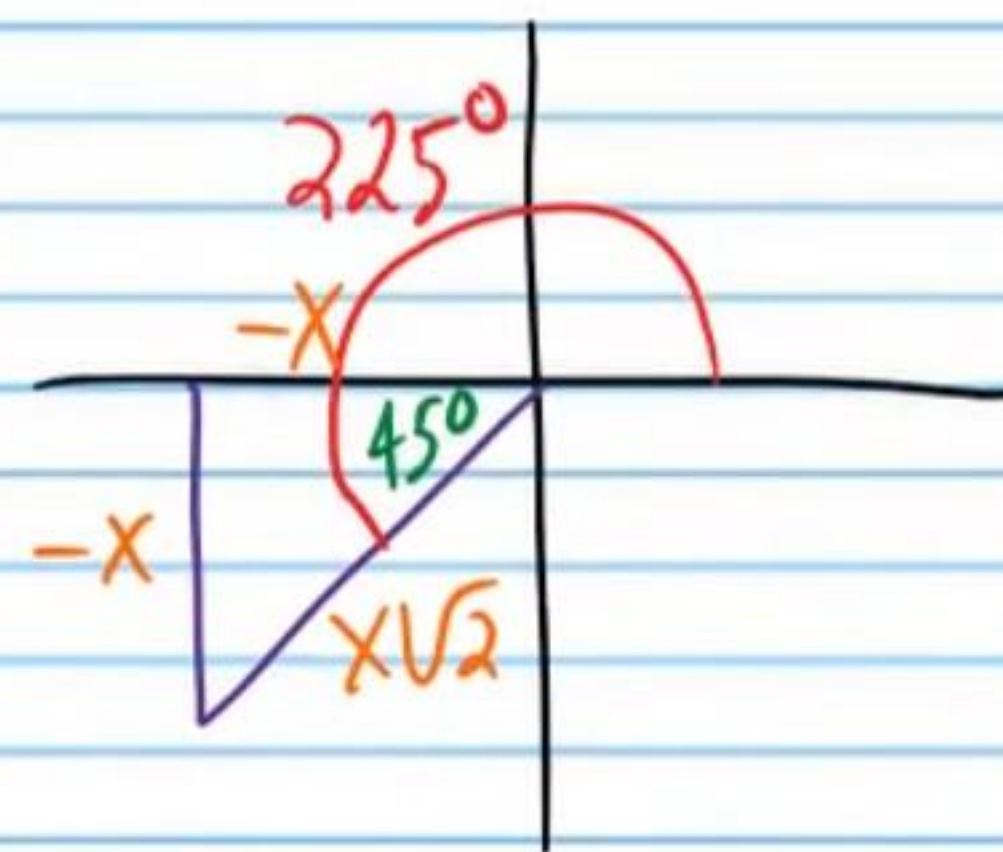


⑭ $\tan^{-1}(\tan 225^\circ)$

I) $\tan 225^\circ$

$$\frac{-x}{-x}$$

$$1$$



II) $\tan^{-1}(\tan 225^\circ)$

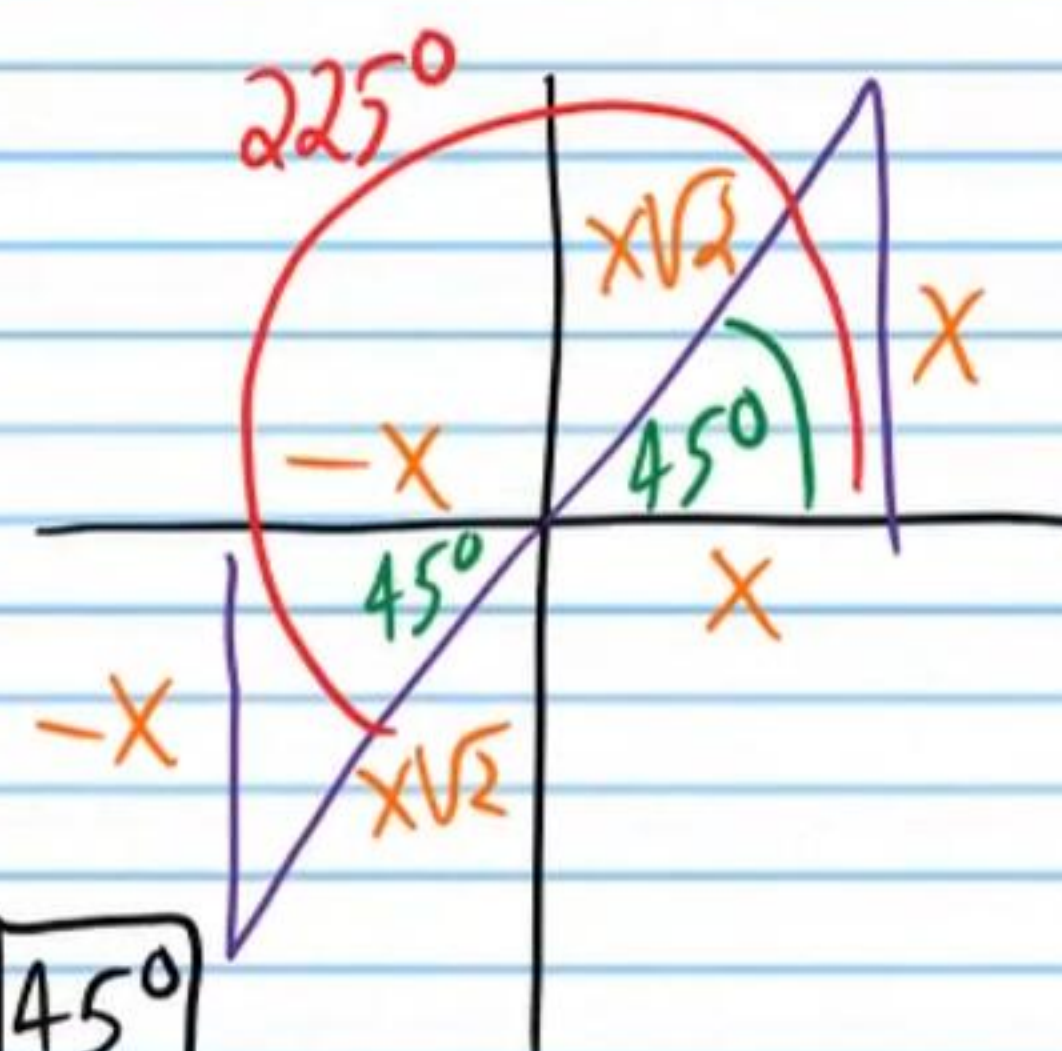
$$\tan^{-1} 1$$

$$\theta = \tan^{-1} 1$$

$$\tan \theta = 1$$

$$\theta = 45^\circ, 225^\circ$$

$$\tan^{-1}(\tan 225^\circ) = \boxed{45^\circ}$$



• (15) $\sin^{-1}(\sin 30^\circ)$

$\boxed{30^\circ}$

(17) $\tan^{-1}(\tan 45^\circ)$

$\boxed{45^\circ}$

(18) $\csc^{-1}(\csc 225^\circ)$

I) $\csc 225^\circ$

$\frac{X\sqrt{2}}{-X}$

$-\sqrt{2}$

II) $\csc^{-1}(\csc 225^\circ)$

$\csc^{-1}(-\sqrt{2})$

$\theta = \csc^{-1}(-\sqrt{2})$

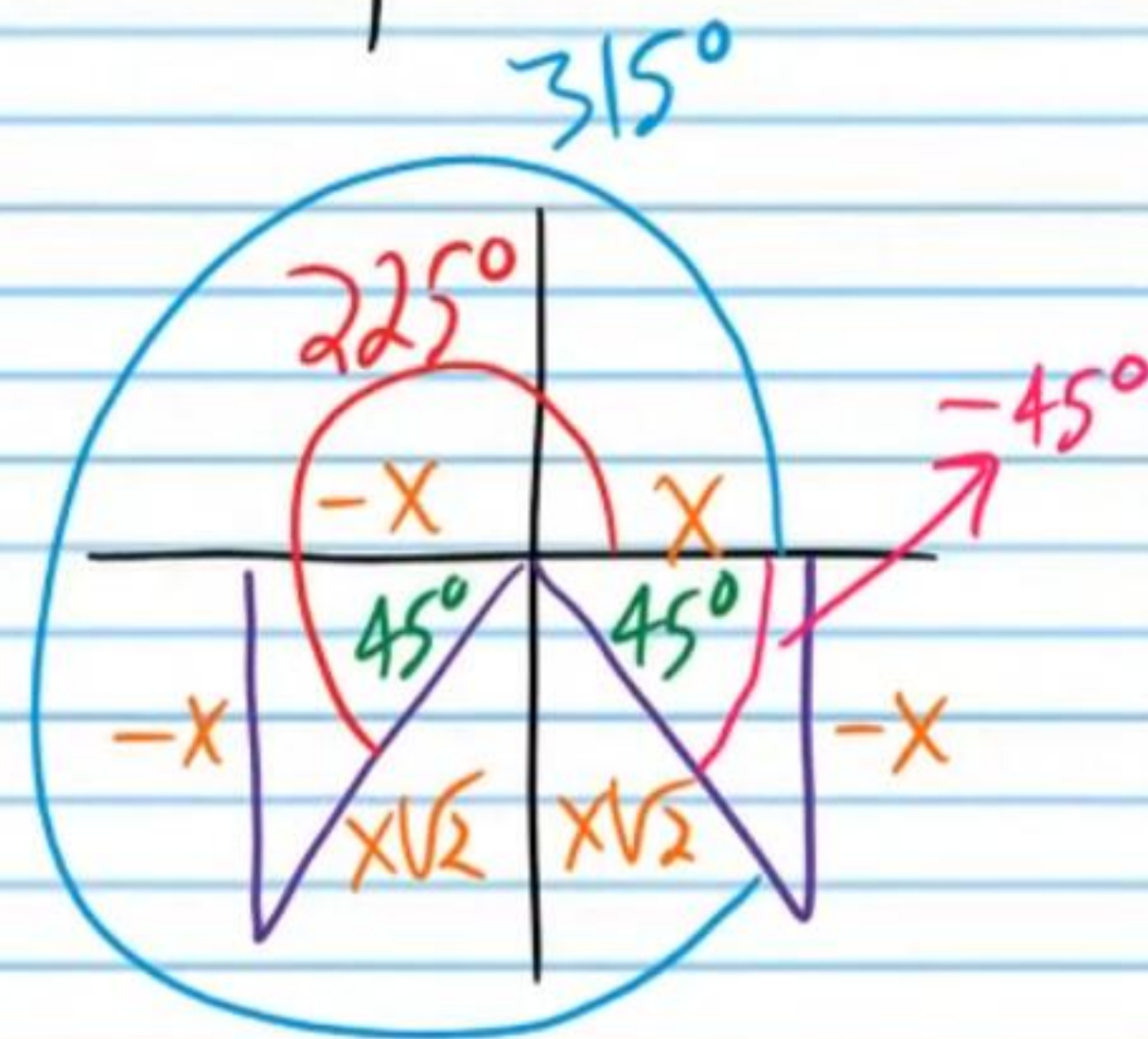
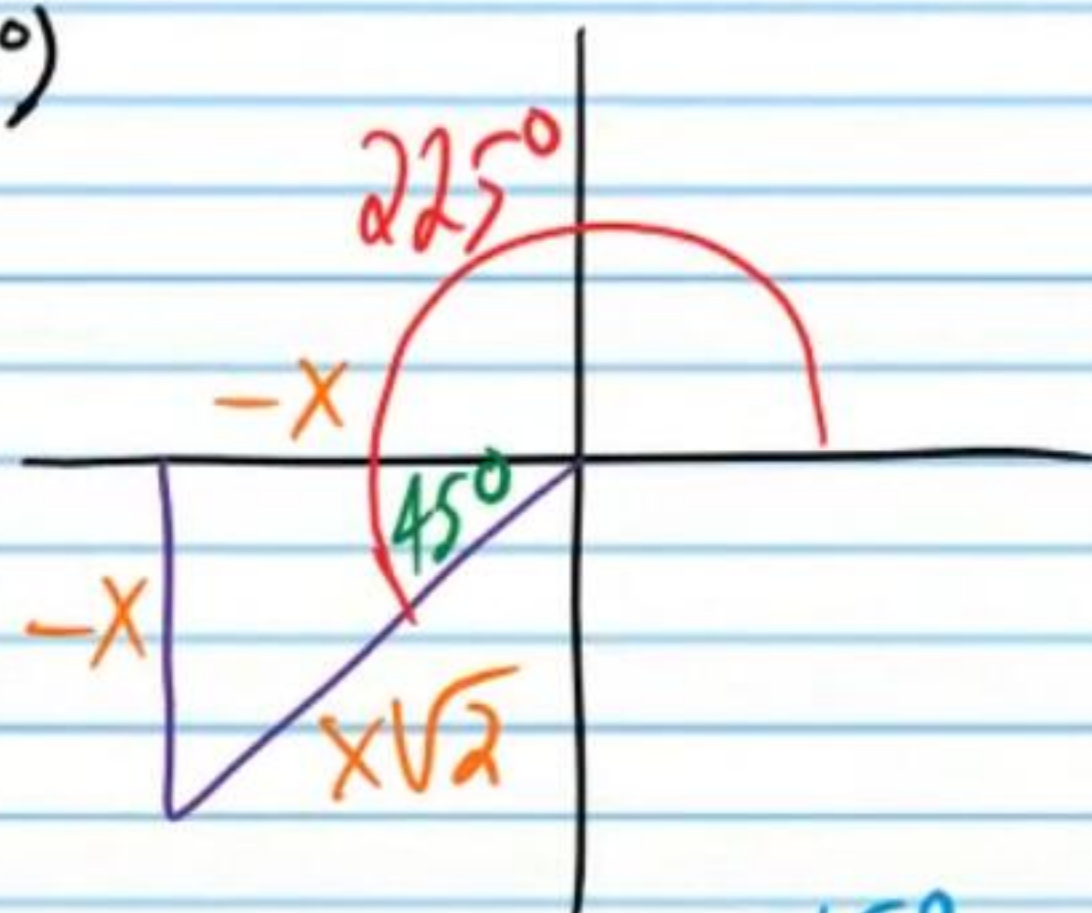
$\csc \theta = -\sqrt{2}$

$\theta = 225^\circ, 315^\circ$

$\csc^{-1}(\csc 225^\circ) = \boxed{-45^\circ}$

(16) $\sec^{-1}(\sec 150^\circ)$

$\boxed{150^\circ}$



(19) $\cos^{-1}(\cos 300^\circ)$

I) $\cos 300^\circ$

$\frac{X}{2X}$

$\frac{1}{2}$

II) $\cos^{-1}(\cos 300^\circ)$

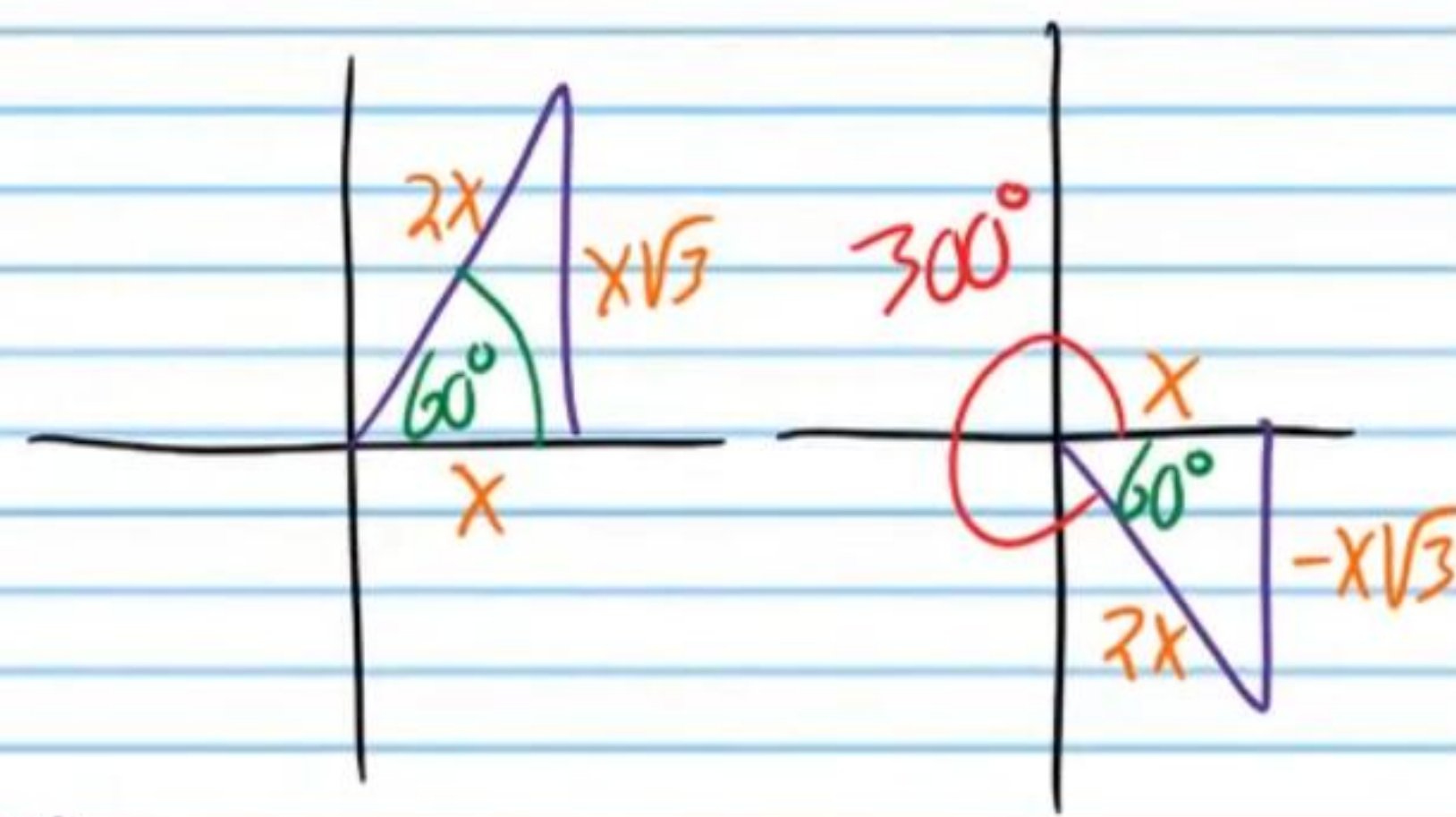
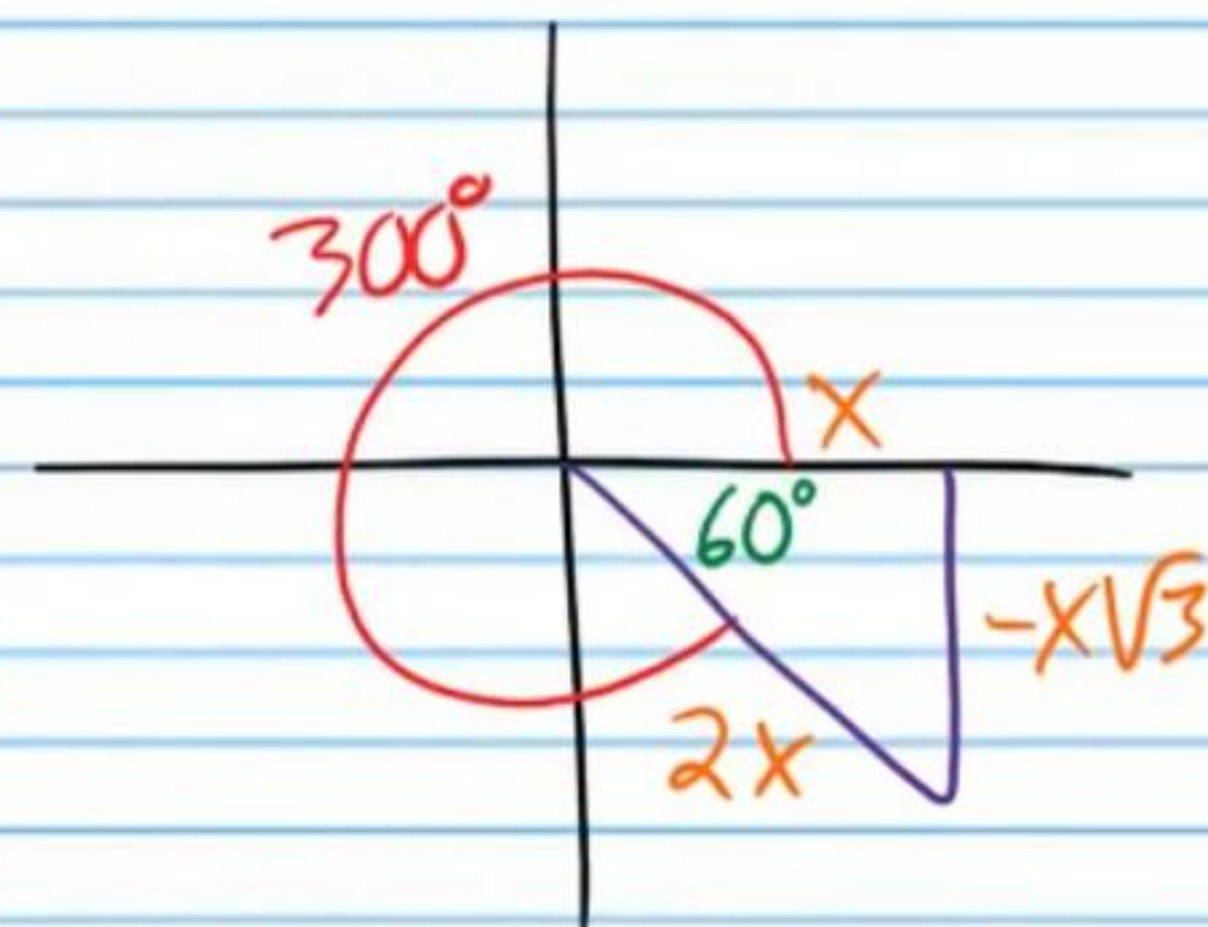
$\cos^{-1}\frac{1}{2}$

$\theta = \cos^{-1}\frac{1}{2}$

$\cos \theta = \frac{1}{2}$

$\theta = 60^\circ, 300^\circ$

$\cos^{-1}(\cos 300^\circ) = \boxed{60^\circ}$

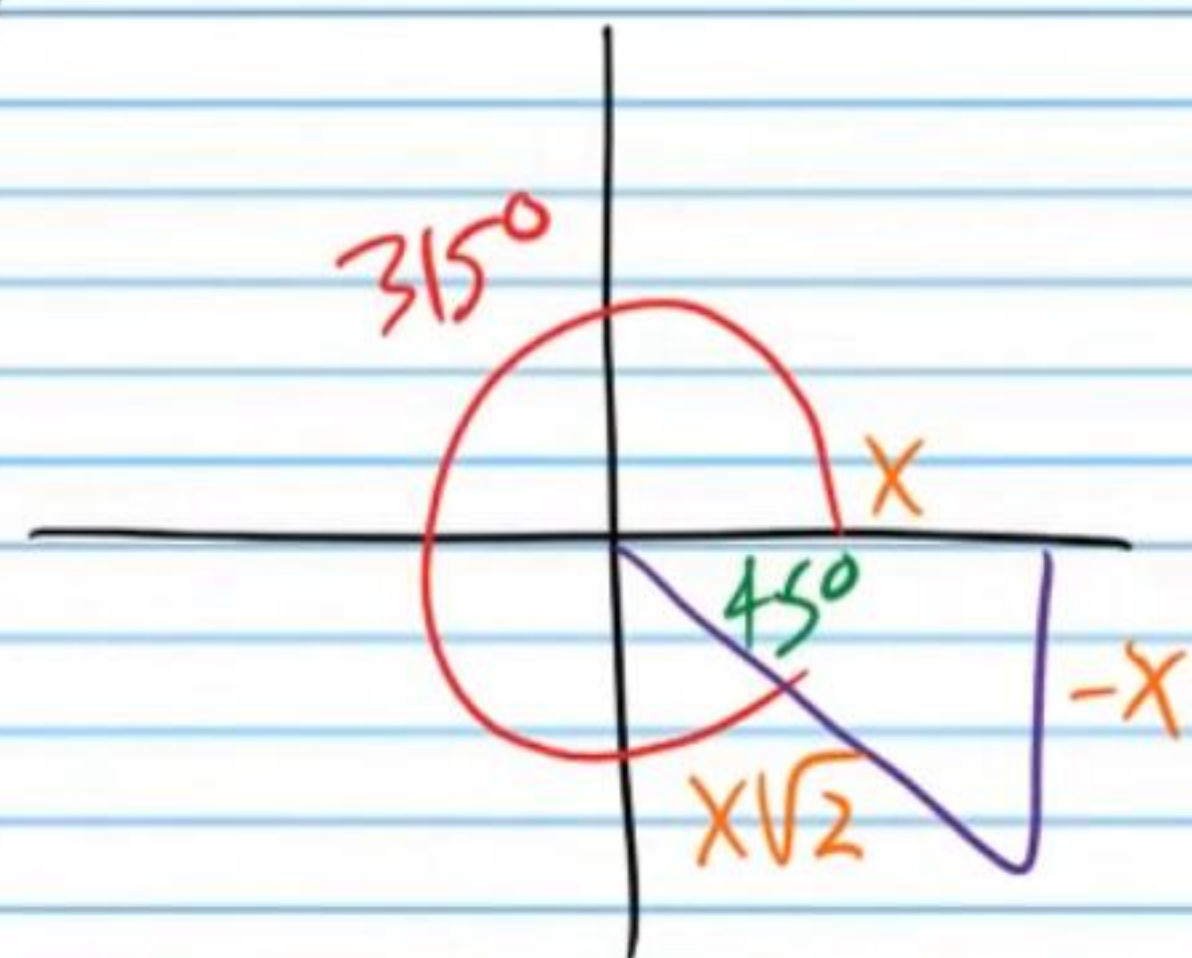


② $\cot^{-1}(\cot 315^\circ)$

I) $\cot 315^\circ$

$$\frac{x}{-x}$$

$$-1$$



II) $\cot^{-1}(\cot 315^\circ)$

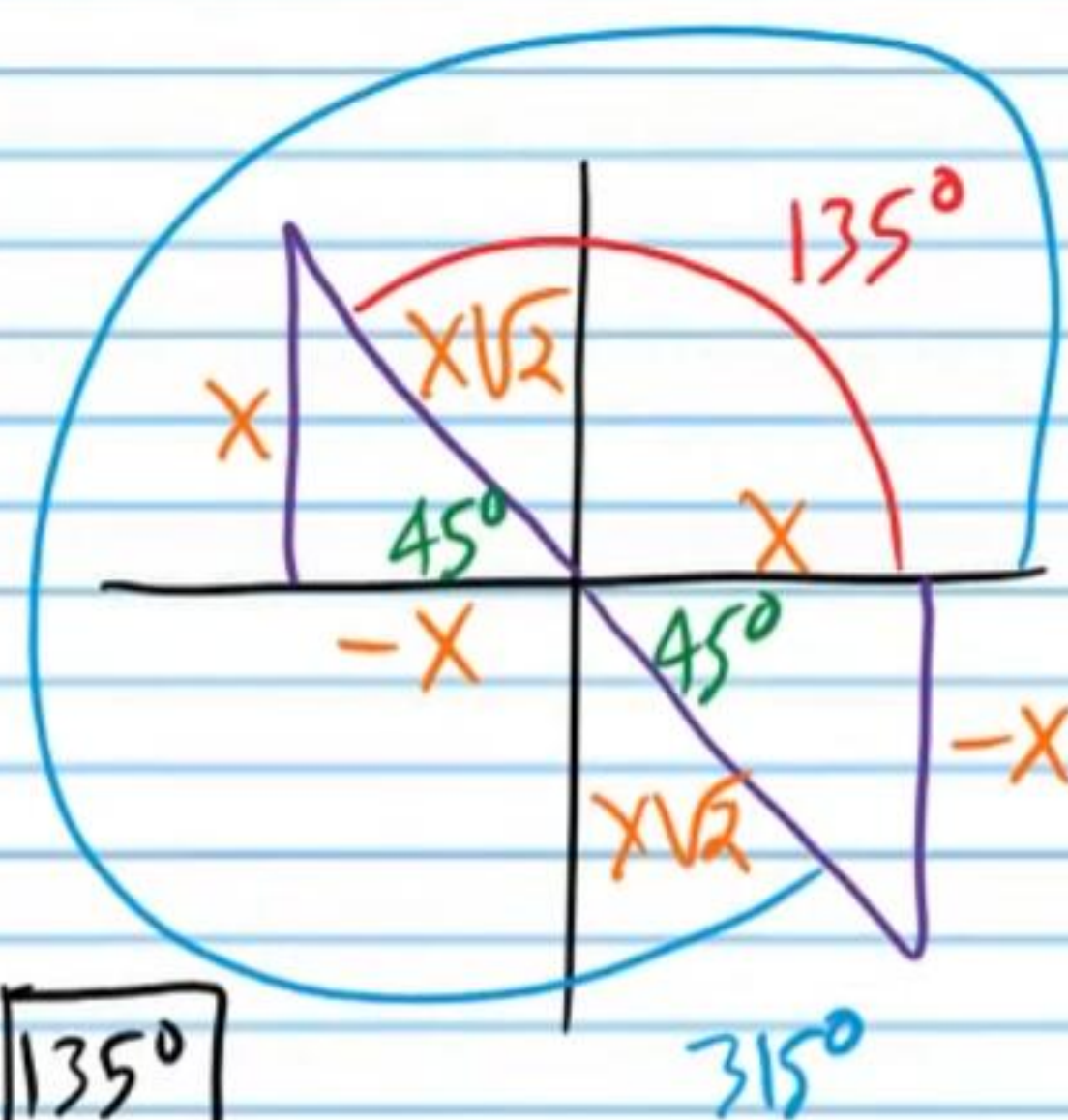
$$\cot^{-1}(-1)$$

$$\theta = \cot^{-1}(-1)$$

$$\cot \theta = -1$$

$$\theta = 135^\circ, 315^\circ$$

$$\cot^{-1}(\cot 315^\circ) = \boxed{135^\circ}$$

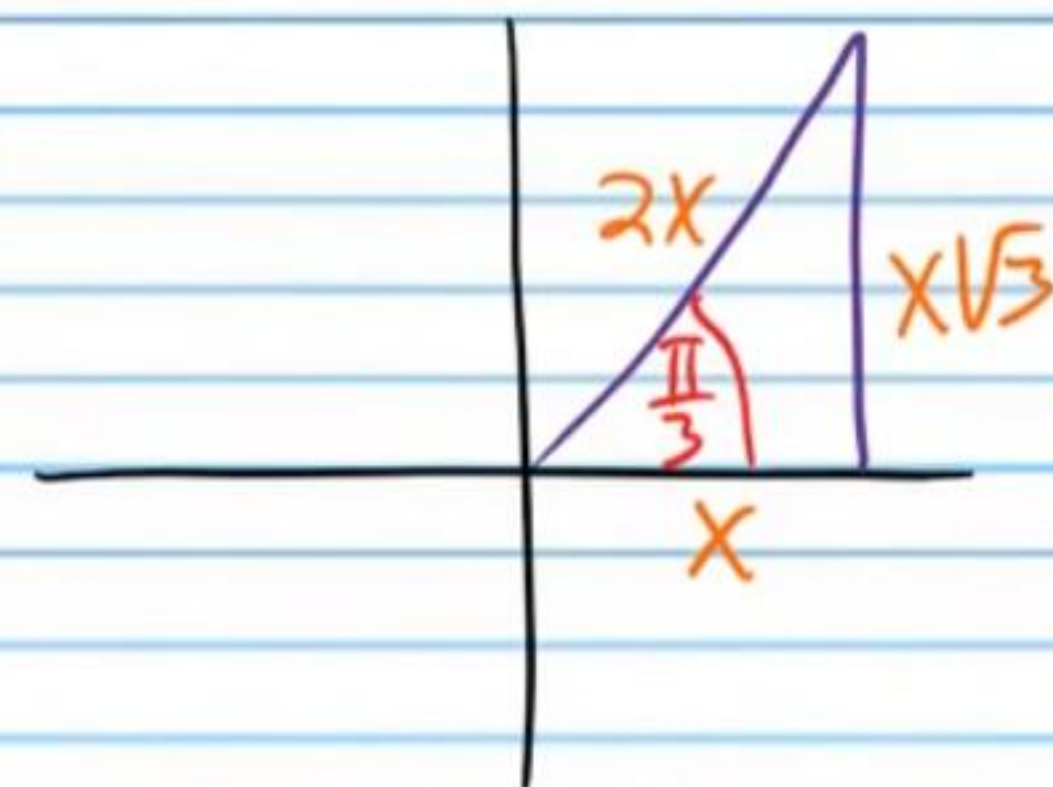


• ② $\cos^{-1}(\sin \frac{\pi}{3})$

I) $\sin \frac{\pi}{3}$

$$\frac{x\sqrt{3}}{2x}$$

$$\frac{\sqrt{3}}{2}$$



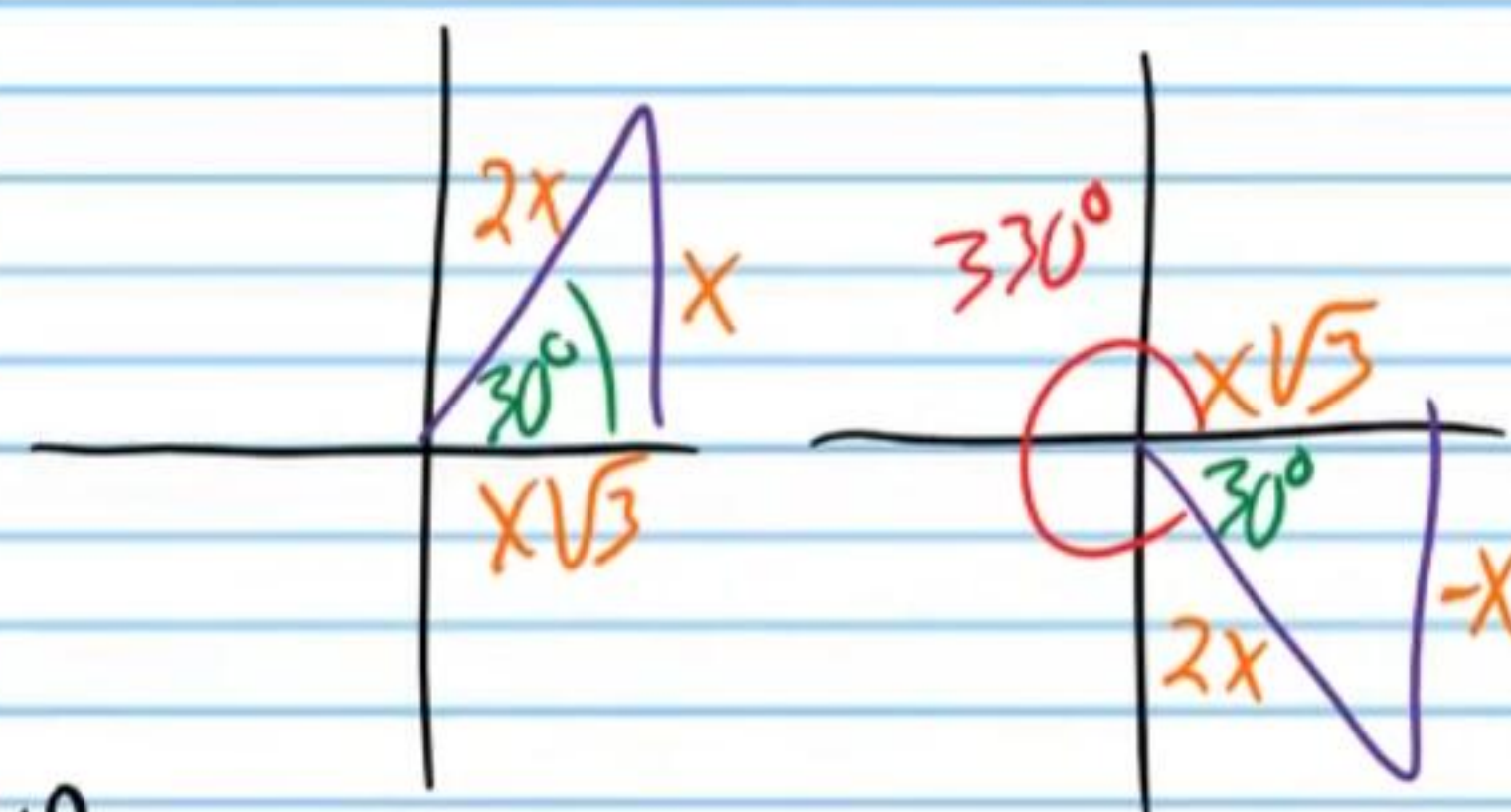
II) $\cos^{-1}(\sin \frac{\pi}{3})$

$$\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$\theta = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$\cos \theta = \frac{\sqrt{3}}{2}$$

$$\theta = 30^\circ, 330^\circ$$



$$\cos^{-1}(\sin \frac{\pi}{3}) = 30^\circ \times \frac{\pi}{180^\circ} = \frac{30\pi}{180} = \boxed{\frac{\pi}{6}}$$

②② $\cot^{-1}(\tan(-\frac{5\pi}{3}))$

I) $\tan(-\frac{5\pi}{3})$

$$\frac{x\sqrt{3}}{x}$$

$$\sqrt{3}$$

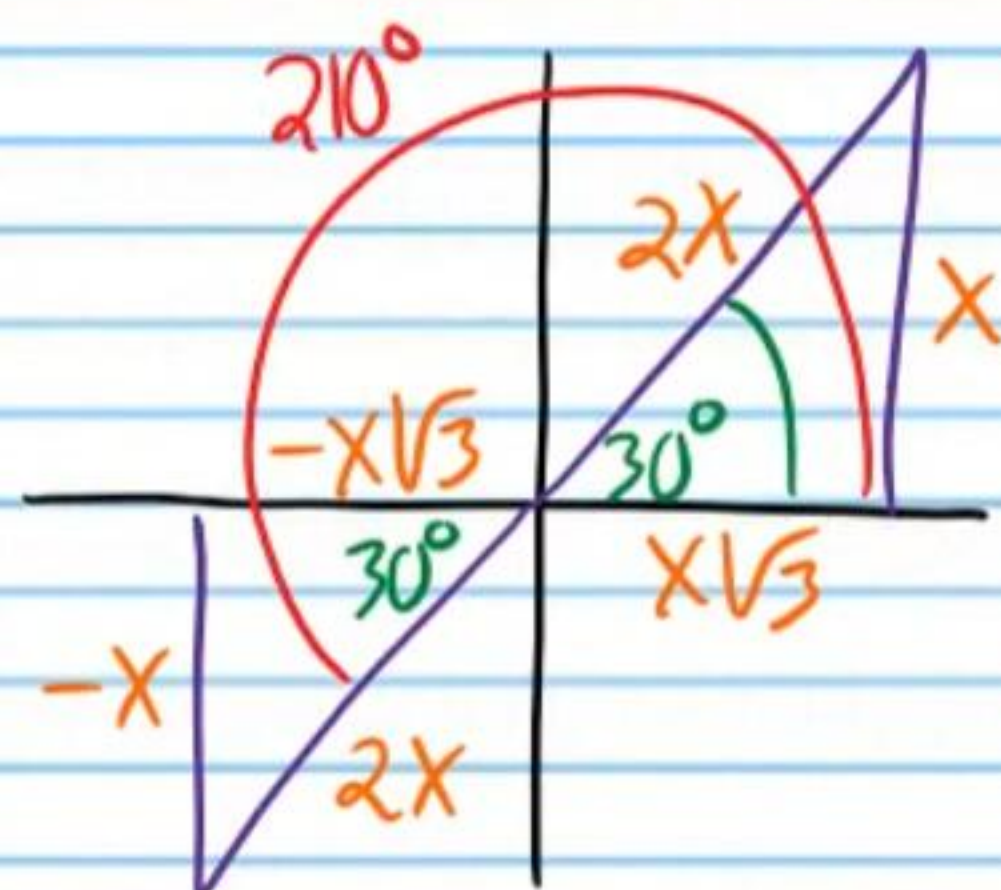
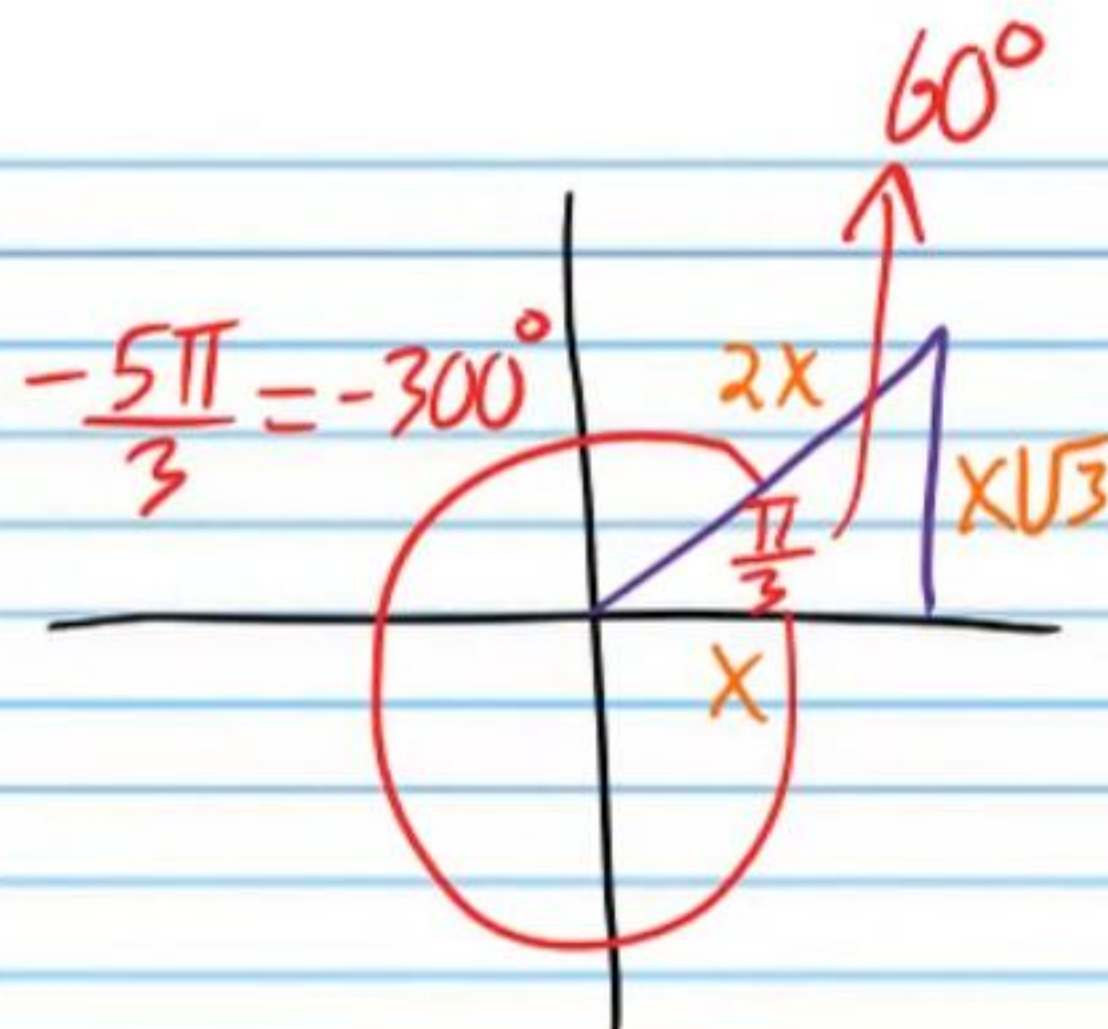
II) $\cot^{-1}(\tan(-\frac{5\pi}{3}))$

$$\cot^{-1} \sqrt{3}$$

$$\theta = \cot^{-1} \sqrt{3}$$

$$\cot \theta = \sqrt{3}$$

$$\theta = 30^\circ, 210^\circ$$



$$\cot^{-1}(\tan(-\frac{5\pi}{3})) = 30^\circ \times \frac{\pi}{180^\circ} = \frac{30\pi}{180} = \boxed{\frac{\pi}{6}}$$

②③ $\cot^{-1}(\cot \frac{\pi}{4})$

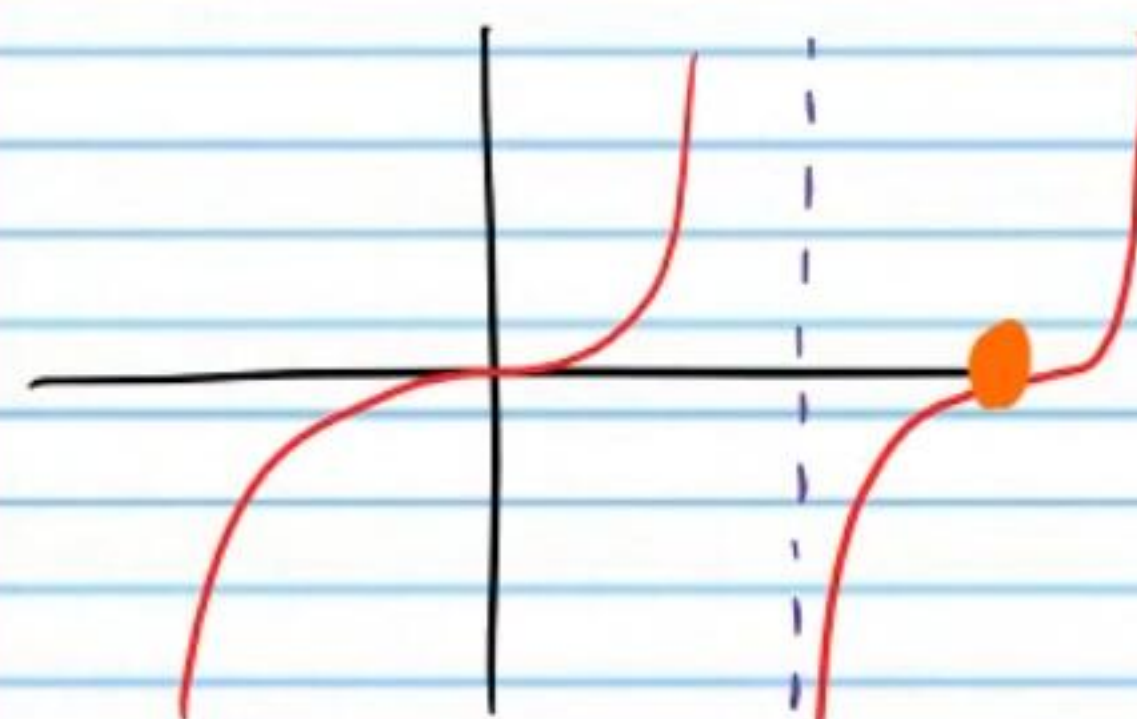
$$\boxed{\frac{\pi}{4}}$$

②④ $\sec^{-1}(\sec \frac{\pi}{6})$

$$\boxed{\frac{\pi}{6}}$$

②⑤ $\tan^{-1}(\tan \pi)$

I) $\tan \pi = 0$



II) $\tan^{-1}(\tan \pi)$

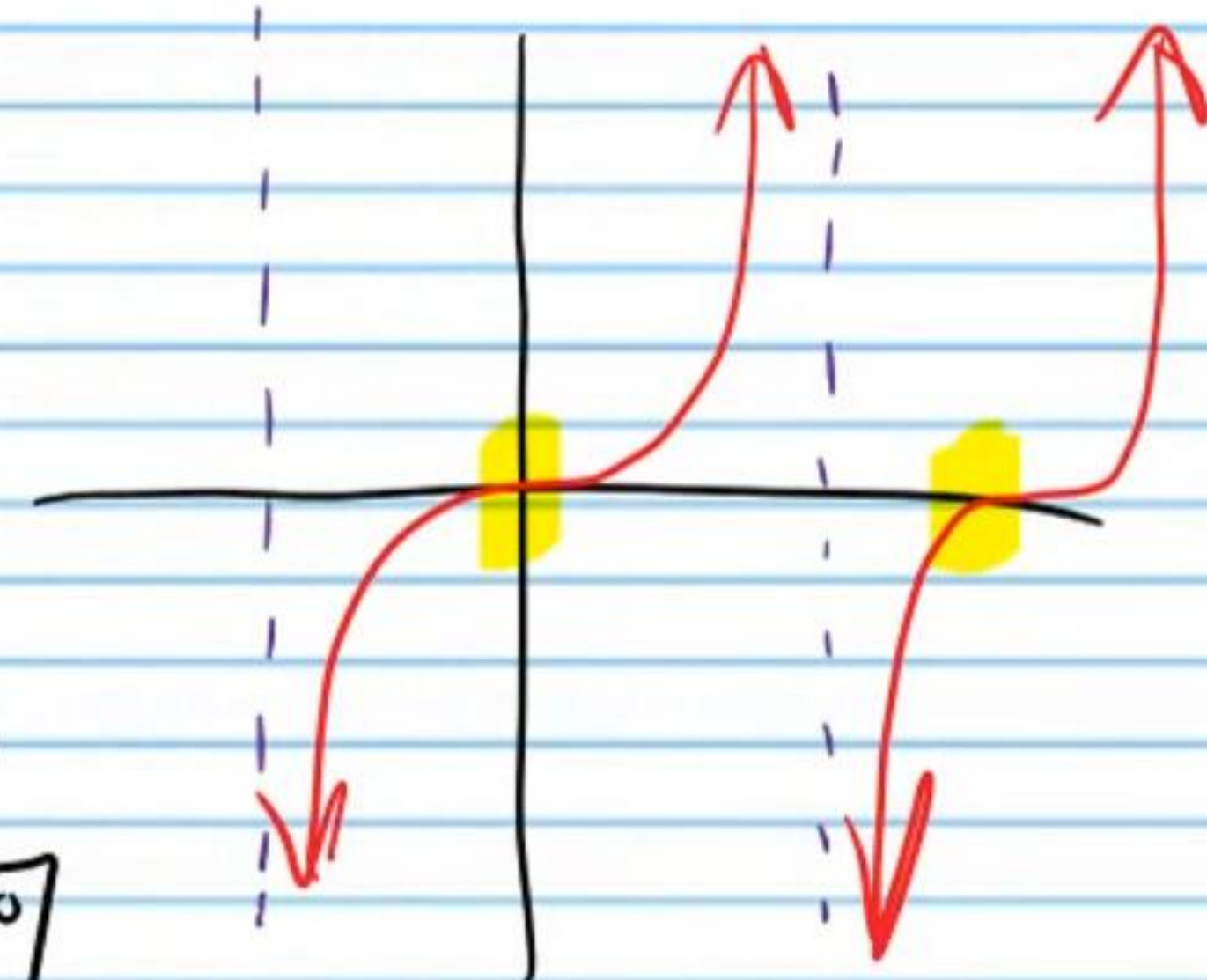
$$\tan^{-1}(0)$$

$$\theta = \tan^{-1}(0)$$

$$\tan \theta = 0$$

$$\theta = 0^\circ, 180^\circ, 360^\circ$$

$$\tan^{-1}(\tan \pi) = \boxed{0^\circ}$$



②⑥ $\csc^{-1}\left(\csc\left(-\frac{3\pi}{4}\right)\right)$

I) $\csc\left(-\frac{3\pi}{4}\right)$

$$\frac{x\sqrt{2}}{-x}$$

$$-\sqrt{2}$$

II) $\csc^{-1}\left(\csc\left(-\frac{3\pi}{4}\right)\right)$

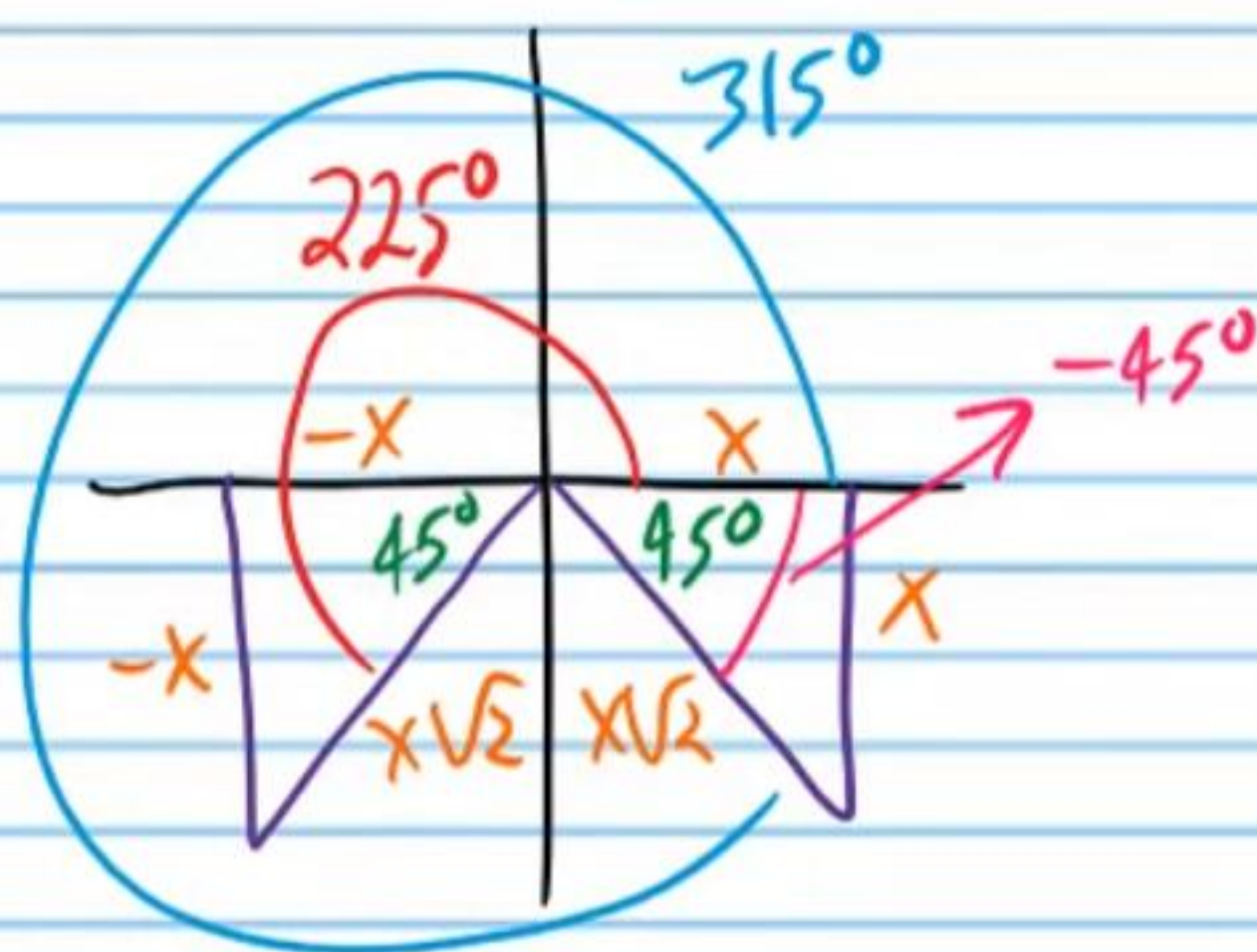
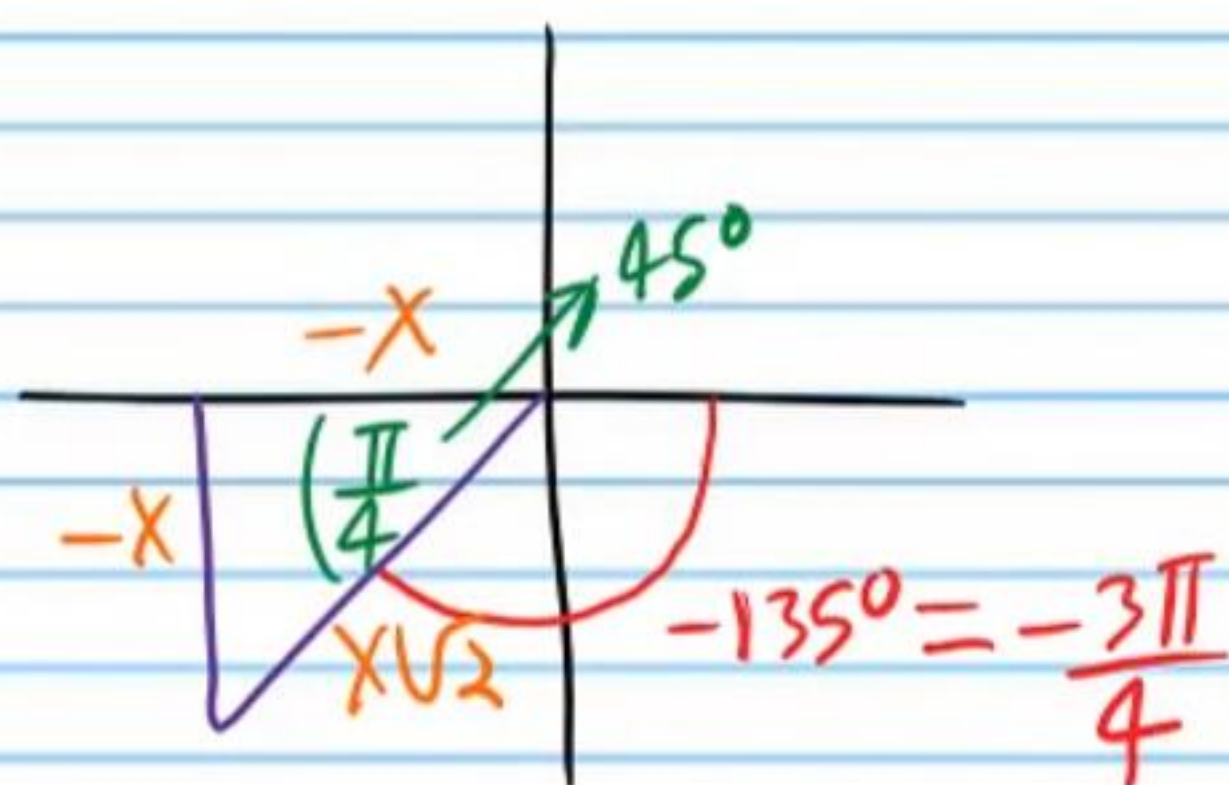
$$\csc^{-1}(-\sqrt{2})$$

$$\theta = \csc^{-1}(-\sqrt{2})$$

$$\csc\theta = -\sqrt{2}$$

$$\theta = 225^\circ, 315^\circ$$

$$\theta = -45^\circ$$



$$\csc^{-1}\left(\csc\left(-\frac{3\pi}{4}\right)\right) = -45^\circ \times \frac{\pi}{180^\circ} = -\frac{45\pi}{180} = \boxed{-\frac{\pi}{4}}$$

• ②⑦ $\sec^{-1}\left(\csc\frac{\pi}{6}\right)$

I) $\csc\frac{\pi}{6}$

$$\frac{2x}{x}$$

$$2$$

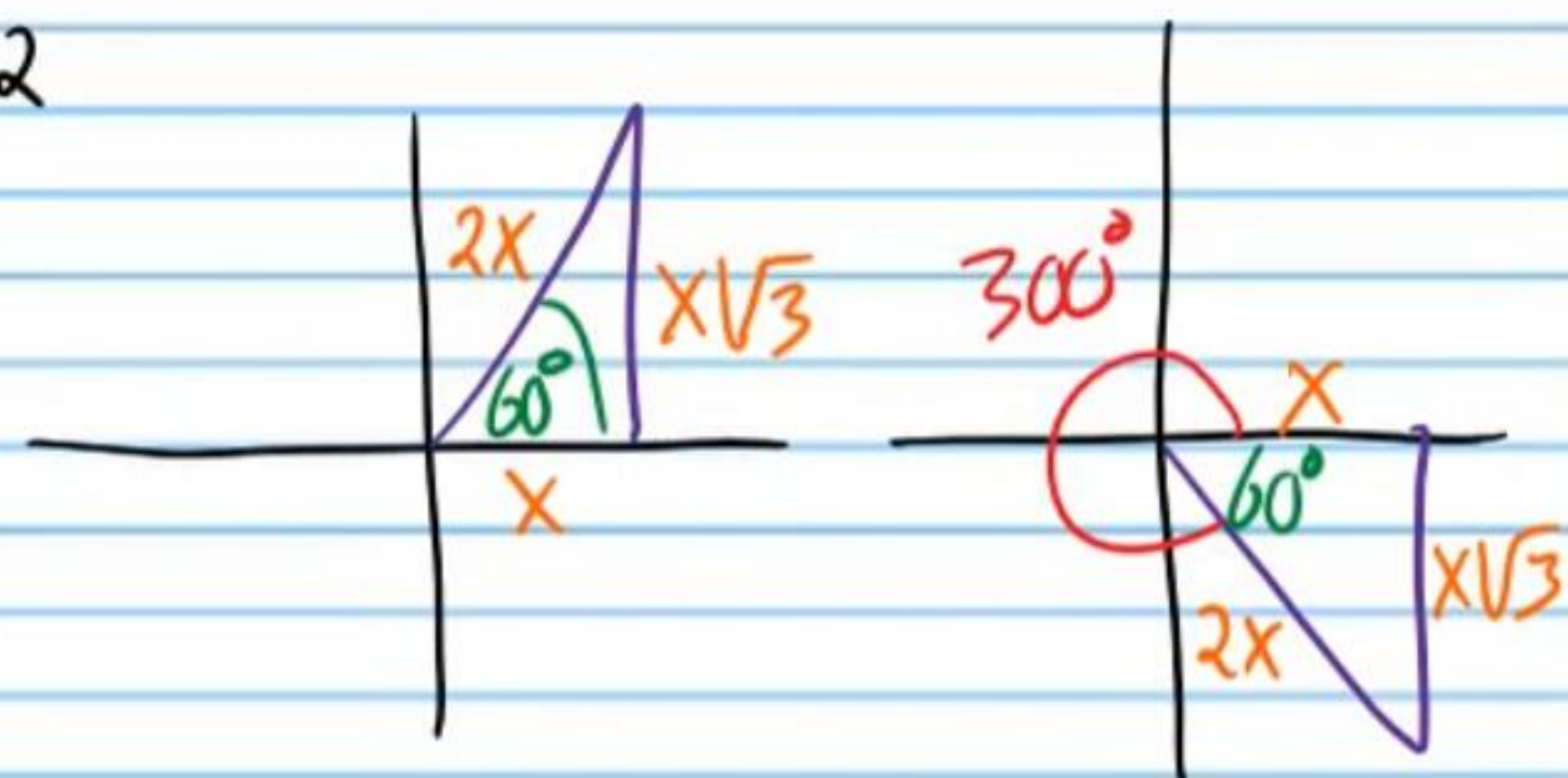
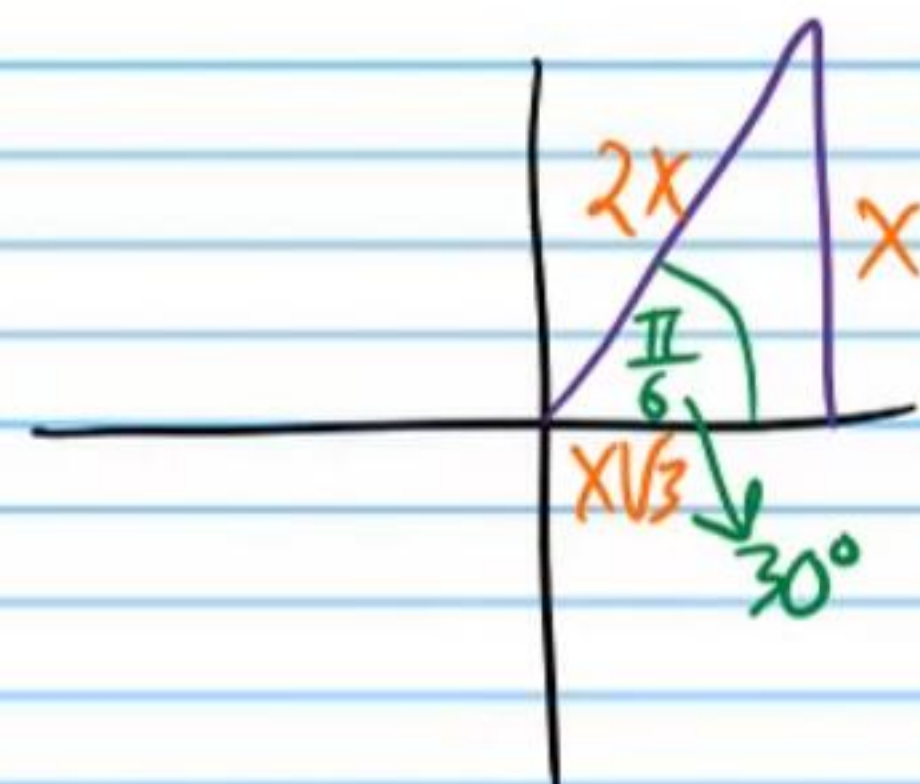
II) $\sec^{-1}\left(\csc\frac{\pi}{6}\right)$

$$\sec^{-1}2$$

$$\theta = \sec^{-1}2$$

$$\sec\theta = 2$$

$$\theta = 60^\circ, 300^\circ$$



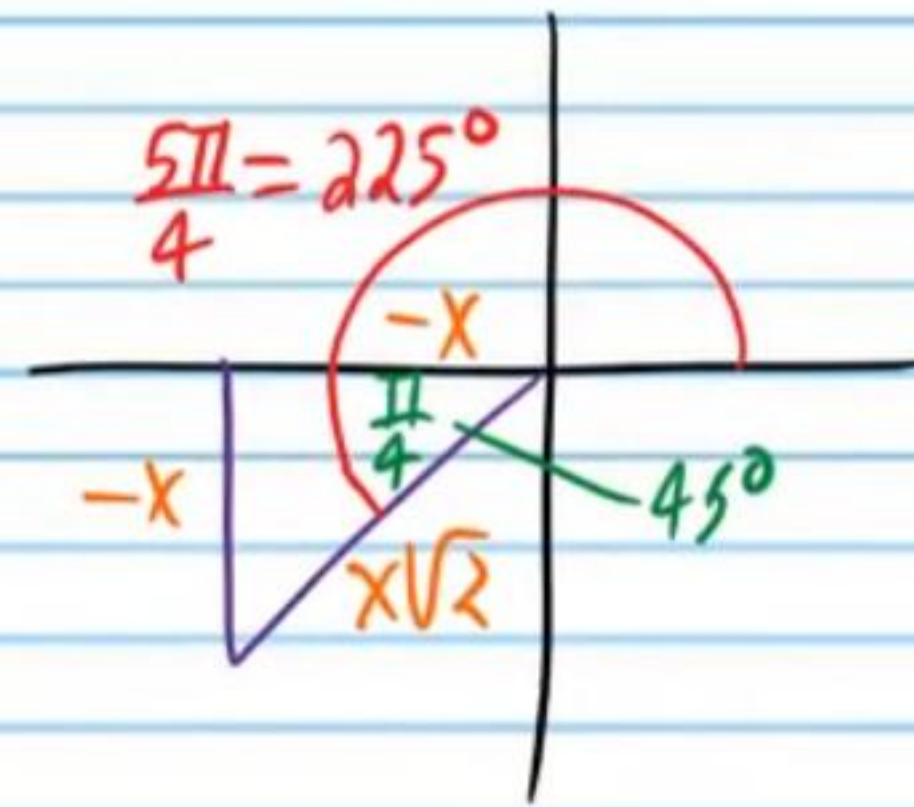
$$\sec^{-1}\left(\csc\frac{\pi}{6}\right) = 60^\circ \times \frac{\pi}{180^\circ} = \frac{60\pi}{180} = \boxed{\frac{\pi}{3}}$$

②⑧ $\csc^{-1}(\sec \frac{5\pi}{4})$

I) $\sec(\frac{5\pi}{4})$

$$\frac{x\sqrt{2}}{-x}$$

$$-\sqrt{2}$$



II) $\csc^{-1}(\sec \frac{5\pi}{4})$

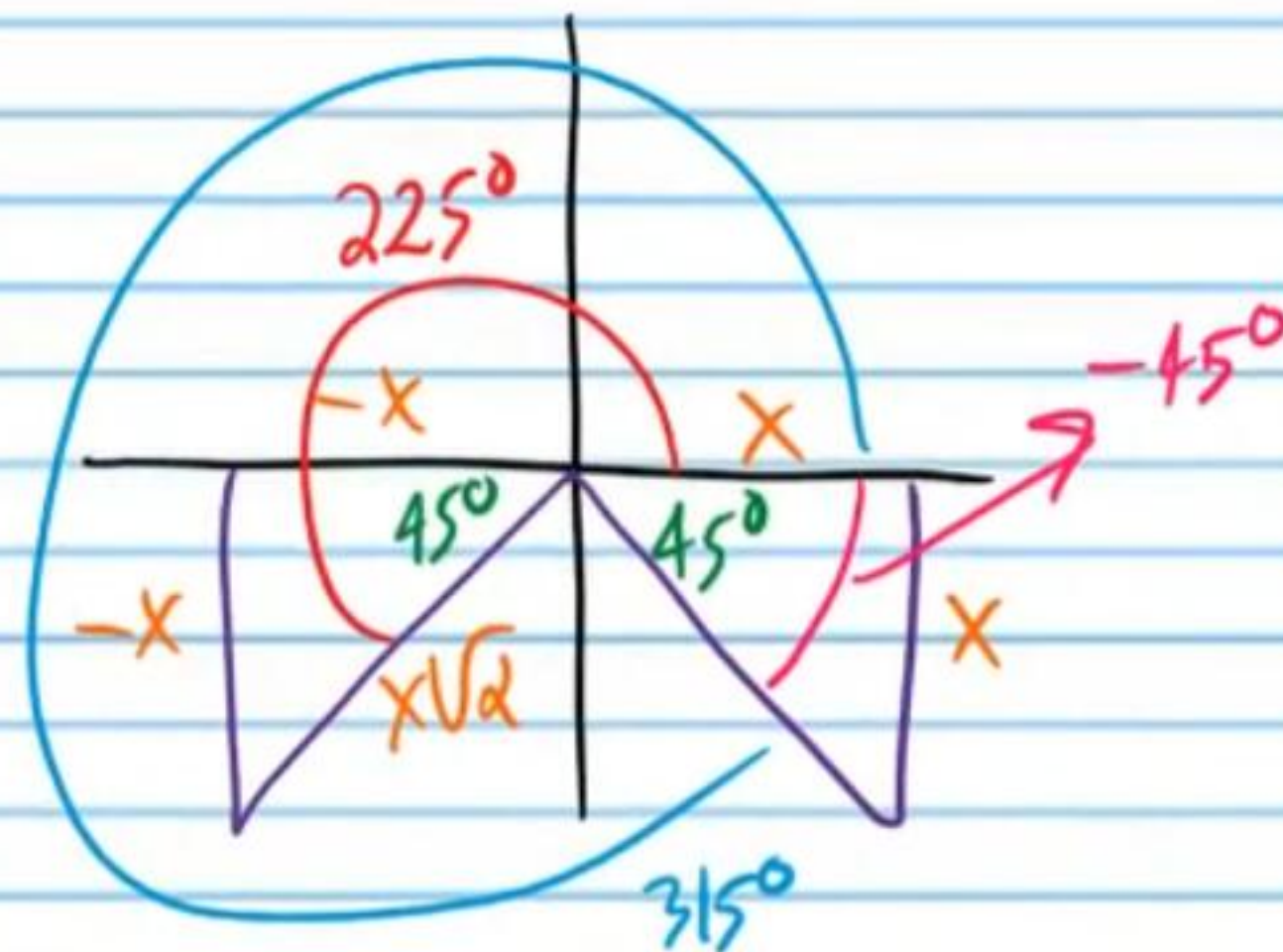
$$\csc^{-1}(-\sqrt{2})$$

$$\theta = \csc^{-1}(-\sqrt{2})$$

$$\csc \theta = -\sqrt{2}$$

$$\theta = 225^\circ, 315^\circ$$

$$\theta = -45^\circ$$



$$\csc^{-1}(\sec \frac{5\pi}{4}) = -45^\circ \times \frac{\pi}{180^\circ} = \boxed{-\frac{\pi}{4}}$$

②⑨ $\tan^{-1}(\tan \frac{\pi}{3})$

$$\boxed{\frac{\pi}{3}}$$

③⑩ $\sin^{-1}(\sin \frac{\pi}{2})$

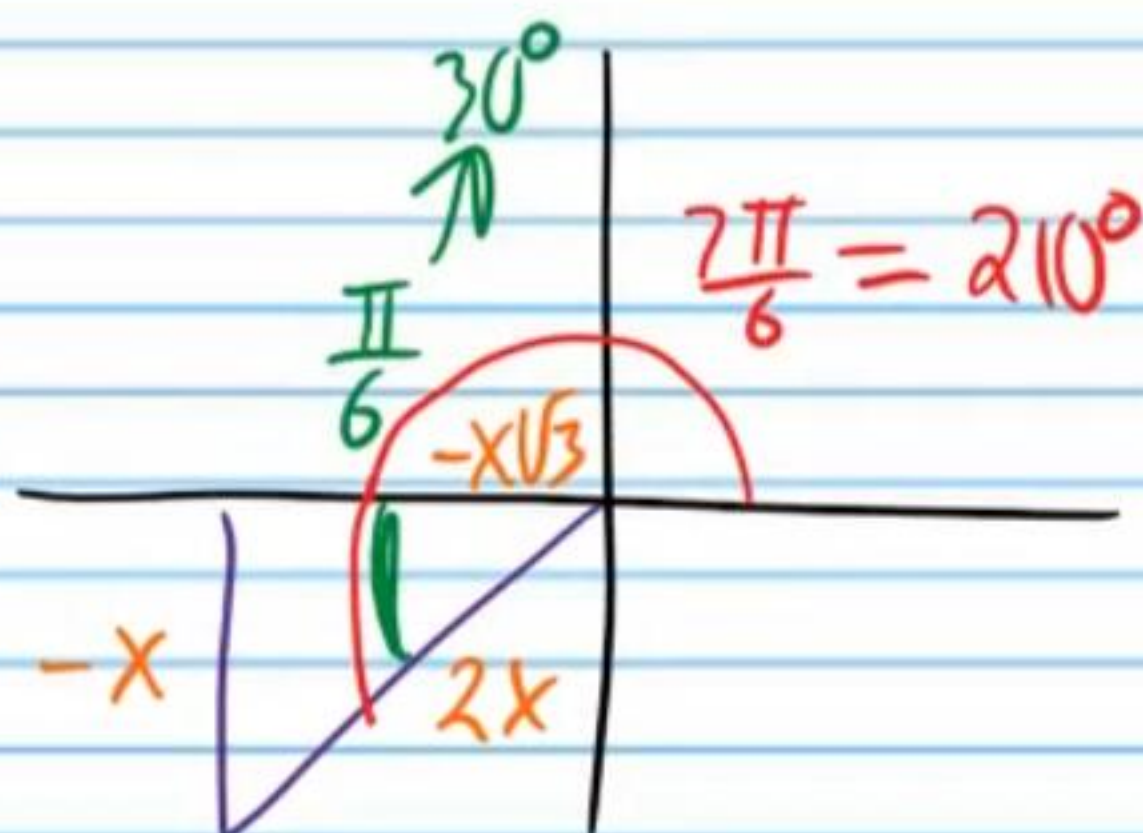
$$\boxed{\frac{\pi}{2}}$$

$$(31) \cos^{-1}(\cos \frac{7\pi}{6})$$

$$I) \cos \frac{7\pi}{6}$$

$$\frac{-x\sqrt{3}}{2x}$$

$$\frac{-\sqrt{3}}{2}$$



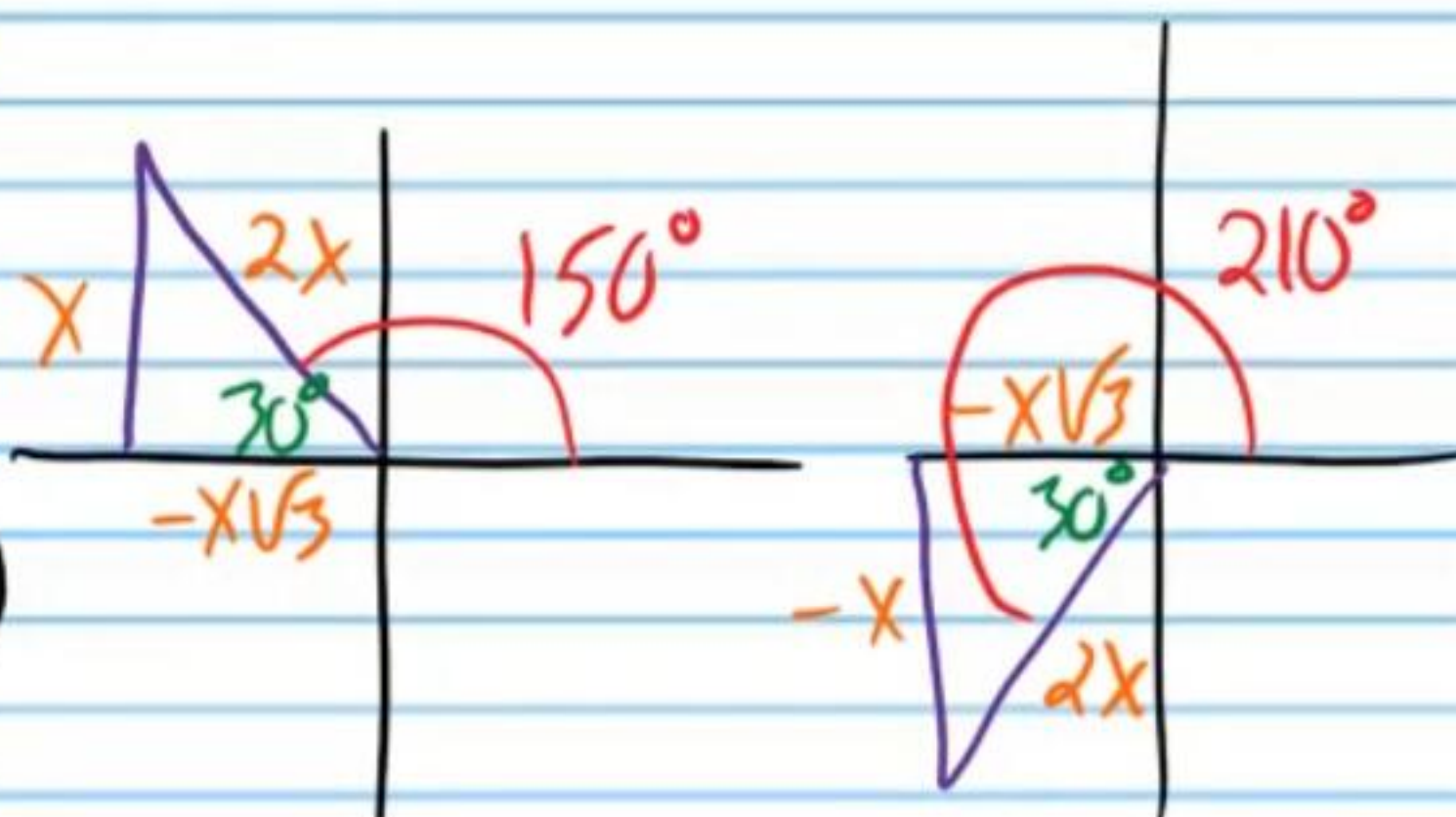
$$II) \cos^{-1}(\cos \frac{7\pi}{6})$$

$$\cos^{-1}(-\frac{\sqrt{3}}{2})$$

$$\theta = \cos^{-1}(-\frac{\sqrt{3}}{2})$$

$$\cos \theta = -\frac{\sqrt{3}}{2}$$

$$\theta = 150^\circ, 210^\circ$$



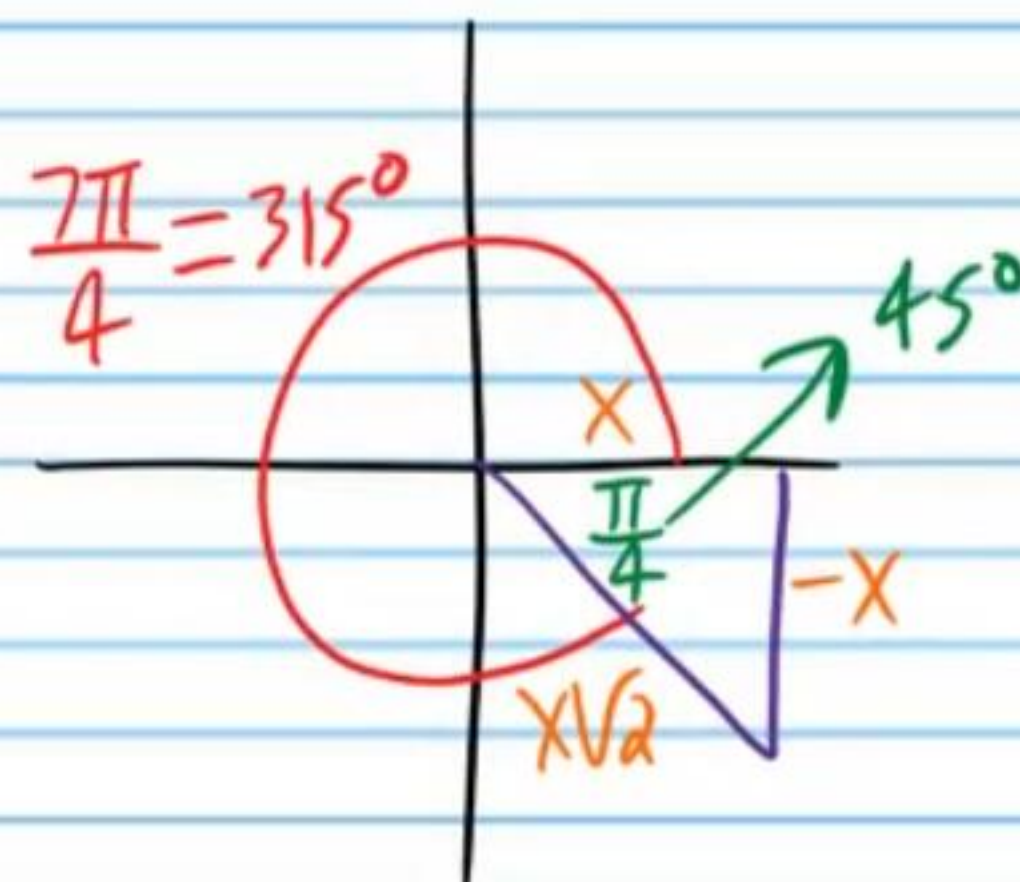
$$\cos^{-1}(\cos \frac{7\pi}{6}) = 150^\circ \times \frac{\pi}{180^\circ} = \frac{150\pi}{180} = \boxed{\frac{5\pi}{6}}$$

$$(32) \cot^{-1}(\cot \frac{7\pi}{4})$$

$$I) \cot \frac{7\pi}{4}$$

$$\frac{x}{-x}$$

$$-1$$



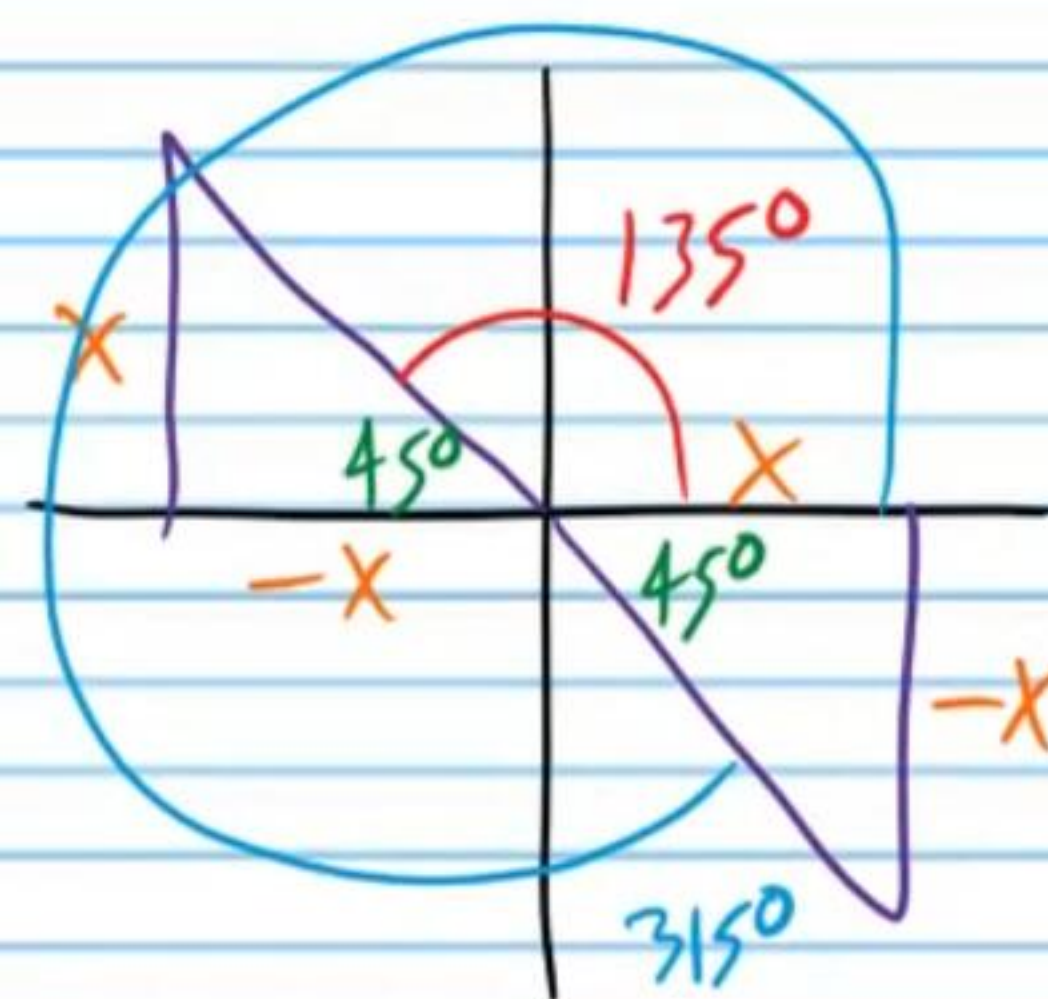
$$II) \cot^{-1}(\cot \frac{7\pi}{4})$$

$$\cot^{-1}(-1)$$

$$\theta = \cot^{-1}(-1)$$

$$\cot \theta = -1$$

$$\theta = 135^\circ, 315^\circ$$



$$\cot^{-1}(\cot \frac{7\pi}{4}) = 135^\circ \times \frac{\pi}{180^\circ} = \frac{135\pi}{180} = \boxed{\frac{3\pi}{4}}$$